

**COMSATS University Islamabad,
Sahiwal Campus**

**Lensify: AI-Powered E-commerce for Glasses & Lenses with
Virtual Try-On**

**A project presented to
COMSATS Institute of Information Technology, Islamabad,
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**In partial fulfilment
of the requirements for the degree of**

Bachelor of Science in Computer Science (2022-2026)

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CERTIFICATE OF APPROVAL

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Executive Summary

The rapid growth of online shopping has changed the way people buy products, including eyewear. However, most existing online eyewear platforms still do not provide a proper way for users to check how glasses will look on their face before buying. Because of this limitation, users often feel unsure while making purchase decisions, which can reduce their confidence and satisfaction. To solve this problem, Lensify is developed as an AI-based virtual eyewear try-on system. The main purpose of this project is to allow users to try different glasses virtually using their camera so they can clearly see how a product looks on their face in real time. This reduces the gap between online shopping and physical experience. Lensify also includes an AI recommendation system that suggests suitable eyewear based on user preferences and facial appearance. This helps users find better matching products without manually searching through many items. Along with this, the system provides an AI chatbot assistant that guides users, answers questions, and helps them understand how to use different features of the system. This improves user interaction and makes the platform more intelligent and supportive. Overall, Lensify combines virtual try-on technology, AI recommendations, and chatbot support into one platform. The aim of this system is to make online eyewear shopping more interactive, accurate, and easy for users, so they can make better and more confident purchase decisions.

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Abbreviations

The abbreviations used throughout this document are defined as follows:

Abbreviation	Definition
AI	Artificial Intelligence
API	Application Programming Interface
AR	Augmented Reality
BCS	Bachelor of Science in Computer Science
CI/CD	Continuous Integration/Continuous Deployment
FR	Functional Requirement
ML	Machine Learning
NFR	Non-Functional Requirement
PC	Personal Computer
SRS	Software Requirement Specification
UI	User Interface
UX	User Experience
WS	WebSocket

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Chapter 1

Introduction

1 Introduction

This chapter presents an overview of the Lensify system, outlining the background, purpose, and key features of the project. It introduces the problem addressed by the system, highlights the proposed solution, and briefly describes the technologies and methodologies used to develop the platform. Lensify is an AI-powered web-based e-commerce platform developed to enhance the online shopping experience for glasses and lenses [1]. Traditional eyewear shopping platforms often lack user confidence because customers cannot physically try products before purchasing. This limitation frequently leads to incorrect buying decisions and reduced customer satisfaction. To address this issue, Lensify integrates advanced technologies such as Virtual Try-On (VTO), face-shape detection, and AI-based recommendation systems [2]. The platform allows users to browse a wide range of eyewear products, virtually try frames using a webcam or uploaded image, and receive personalized suggestions based on their facial features. In addition, users can enter and manage their prescription details, customize lenses, and complete secure transactions through integrated payment methods. The system also includes an admin module for managing products, orders, and user data efficiently.

1.1 Brief

Lensify is an AI-powered e-commerce platform developed to improve the online shopping experience for glasses and lenses. The project focuses on solving the common issue of user uncertainty in selecting eyewear by introducing a Virtual Try-On feature and intelligent recommendations based on face shape. The system enables users to browse products, try frames in real time using a webcam or image, customize lenses with prescription details, and complete secure online purchases. The outcome of this project is a fully functional, web-based application that provides an interactive and personalized shopping experience. It successfully integrates key modules such as user authentication, product management, Virtual Try-On, AI recommendation engine, cart and order processing, and payment integration [4]. The system also includes an admin dashboard for efficient management of products and order.

The development of Lensify utilizes modern tools and technologies including React.js for frontend development, Node.js and Express.js for backend services, MongoDB Atlas for database

management, and TensorFlow.js with Media Pipe for AI-based face detection and Virtual Try-On functionality. The project follows an Object-Oriented Design approach along with the Incremental development model to ensure modularity, scalability, and continuous improvement. This report is structured to cover all aspects of the system. The initial chapters introduce the project background and requirements, followed by system design and architecture. Subsequent chapters explain data models, algorithms, and implementation details. The final sections discuss testing, limitations, and future enhancements, providing a complete overview of the Lensify system.

1.2 Relevance to Course Modules

The Lensify project is closely related to multiple courses studied during the Bachelor of Computer Science (BCS) program. It integrates concepts from different subjects and applies them in a practical, real-world system.

Firstly, the project reflects the concepts of Software Engineering, where Software Development Life Cycle (SDLC), requirement analysis, system design (SRS, SDD), and testing methodologies are applied. The use of Object-Oriented Design and the Incremental development model also come from this course.

Secondly, Object-Oriented Programming (OOP) plays a key role in designing system components such as User, Product, Cart, and Order. Concepts like classes, inheritance, and modularity are used to build a structured and maintainable system.

The project also utilizes Database Systems knowledge by implementing a NoSQL database using MongoDB Atlas. Concepts like data modeling, indexing, and efficient data retrieval are applied to manage users, products, orders, and transactions.

From the Web Development course, technologies like React.js, Node.js, and Express.js are used to build the frontend and backend of the system. RESTful APIs are implemented to enable communication between different modules.

The Artificial Intelligence and Machine Learning course is reflected in features such as face-shape detection and personalized recommendations. Libraries like TensorFlow.js and Media Pipe are used to implement intelligent functionalities.

Additionally, Human Computer Interaction (HCI) principles are applied to design a user-friendly interface with smooth navigation and intuitive layout.

Lastly, concepts from Computer Networks and Information Security are used in handling secure data transmission, authentication, and payment processing through integrated gateways.

1.3 Project Background

The idea behind the Lensify project originates from the growing trend of online shopping and the limitations faced in the eyewear industry. While e-commerce platforms have made it easy to purchase products online, buying glasses and lenses still remains challenging because customers cannot physically try them before making a decision. This often leads to uncertainty, incorrect choices, and reduced customer satisfaction. To address this issue, Lensify is based on the concept of Virtual Try-On (VTO) and Artificial Intelligence (AI) [5]. Virtual Try-On is a technology that allows users to digitally visualize how a product will look on them. In the case of eyewear, it uses a webcam or uploaded image to detect the user's face and overlay selected frames in real time, giving a realistic preview. This process involves detecting facial landmarks such as eyes, nose, and face boundaries, and then accurately positioning the glasses on those points. Along with VTO, the system uses AI-based face shape detection. This technology analyzes facial features and classifies the face into categories such as oval, round, square, or heart-shaped. Based on this classification, the system recommends frames that best suit the user's face, making the shopping experience more personalized and efficient [7].

The project also incorporates the concept of e-commerce systems, where users can browse products, manage a shopping cart, customize lenses using prescription details, and complete secure transactions through integrated payment gateways.

1.4 Literature Review

This section presents an overview of existing research, technologies, and industry solutions related to AI-based e-commerce and virtual try-on systems used in eyewear platforms. In recent years, virtual try-on (VTO) technology has become an important innovation in online shopping. It helps users visualize products before purchasing, solving one of the biggest problems in e-commerce lack of physical interaction. Research shows that VTO systems use computer vision and augmented reality (AR) to overlay virtual products on a user's face or body in real time, improving

user confidence and reducing product returns. Many studies highlight that virtual try-on systems work by detecting facial features using techniques like face landmark detection, head pose estimation, and 3D modeling. These techniques allow accurate placement of glasses on a user's face through a webcam or image. Recent advancements in deep learning and AI have further improved virtual try-on systems. Modern systems now use machine learning models to analyze user features such as face shape and recommend suitable products automatically. Research also shows that virtual try-on can be divided into 2D image-based and 3D model-based approaches, each having its own advantages in terms of speed and realism. In the commercial market, companies like Warby Parker and Lenskart have already implemented AR-based try-on features to enhance customer experience. These platforms allow users to try glasses virtually using mobile apps or web cameras, making online shopping more interactive and personalized.

However, literature also points out some limitations in existing systems, such as accuracy issues, unrealistic rendering in some cases, and privacy concerns related to facial data collection. Despite these challenges, virtual try-on technology continues to evolve and is widely considered a key feature in modern e-commerce platforms.

Based on these studies, it is clear that integrating AI, AR, and recommendation systems can significantly improve user experience in online eyewear shopping. The proposed system, Lensify, builds on these concepts by combining virtual try-on, face-shape detection, AI-based recommendations, and secure e-commerce features into one complete platform.

1.5 Analysis from Literature Review

The literature review unequivocally validates the need for a system like Lensify, specifically emphasizing that its proposed AI-enhanced features are critical differentiators in the increasingly saturated digital eyewear marketplace. The fundamental challenge identified in current research is the persistent gap between simple transactional e-commerce platforms and truly intelligent, immersive marketplaces. Lensify's response to the integration of an advanced AI and AR engine aligns precisely with modern research favoring sophisticated biometric analysis and real-time visualization over elementary photo filters or static image displays [7].

For instance, while a basic platform might only filter products by a generic category like Men's Frames, Lensify utilizes computer vision to extract embeddings from the user's live facial

geometry. By calculating specific ratios and landmark coordinates, the system generates a high-fidelity compatibility score that maps a user's unique facial structure to specific frame geometries. This ensures a significantly higher success rate for the Try-On session and optimizes the user's journey by narrowing down the catalog to products that are mathematically and aesthetically guaranteed to fit. Furthermore, the integration of automated face-shape classification directly addresses the established issue of choice paralysis and fit uncertainty, leveraging known geometric classification methodologies to provide tailored styling advice. This feature serves as a digital consultant, increasing user confidence and reducing the high return rates that are prevalent in current systems lacking AR-guided selection.

The technological choices made for Lensify reinforce the system's long-term viability and performance. The decision to use the MERN stack provides a robust, event-driven foundation for building a highly responsive application. By utilizing TensorFlow.js and Media Pipe for client-side processing, the system ensures that resource-intensive machine learning calculations such as 468-point facial mesh tracking do not impair the platform's real-time responsiveness. This architecture maintains a seamless user experience during core actions like live frame overlays and product browsing.

1.6 Methodology and Software Lifecycle for This Project

The development of Lensify necessitates a structured yet flexible approach due to the integration of complex Artificial Intelligence (AI) and Augmented Reality (AR) components within a standard e-commerce framework. To manage these technical risks while ensuring the delivery of a high-quality product, the Incremental Model was selected as the Software Development Life Cycle (SDLC) for this project.

1.6.1 Selection of the Incremental Model

The Incremental Model is a process of software development where requirements are broken down into multiple standalone modules of the development cycle. In this model, the software is designed, implemented, and tested incrementally until the product is finished. This model was prioritized over the traditional Waterfall approach because it allows for the delivery of core functionalities early in the process, providing a "working version" of the platform that can be refined through subsequent iterations.

Given that Lensify involves experimental AI features (like 468-point facial mesh tracking), the Incremental model provided the team with the necessary buffer to perfect the computer vision algorithms without stalling the development of the essential e-commerce backend and database infrastructure.

1.6.2 Implementation of Incremental Phases

The lifecycle of Lensify was divided into three primary increments, each resulting in a functional version of the system:

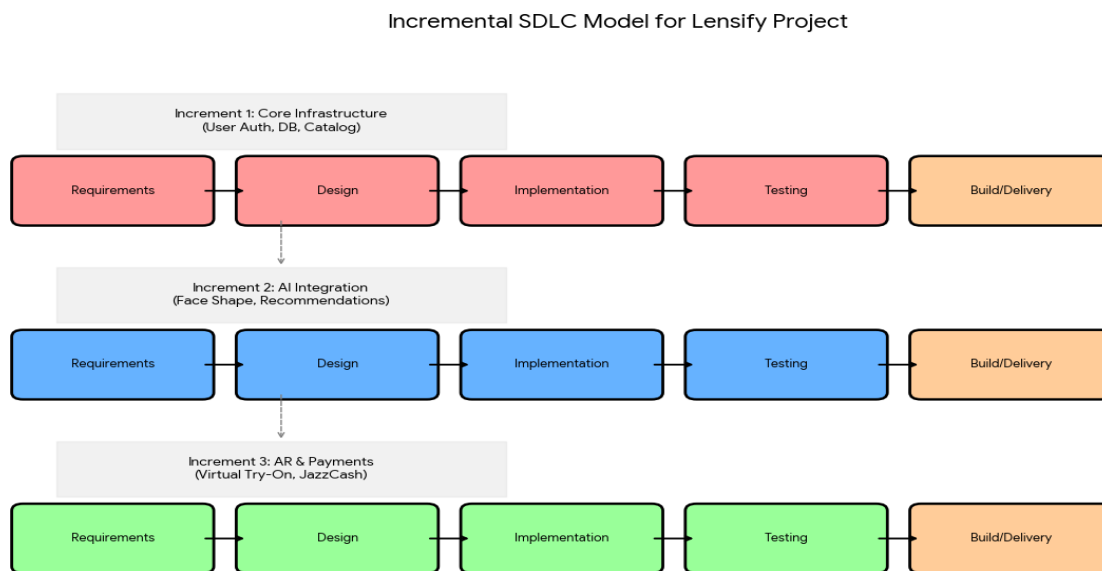


Figure 1.1 Incremental Software Development Life Cycle

Increment 1: Core E-commerce Infrastructure

This phase focused on the "skeleton" of the platform. The primary deliverables included the MERN stack setup, user authentication (Signup/Login), and the product catalog management system. By the end of this increment, the system functioned as a standard online store where users could browse frames and manage their profiles.

Increment 2: AI Integration & Personalization

Once the core platform was stable, the second increment focused on the Intelligence layer. This involved the integration of Media Pipe for face shape detection and the development of the AI-driven recommendation engine. The AI Chatbot was also implemented during this phase to assist users in navigating the catalog based on their detected facial geometry.

Increment 3: AR Virtual Try-On & Final Deployment

The final and most complex increment involved the Augmented Reality layer. This phase focused on real-time rendering, ensuring that the 3D or 2D frame models accurately mapped to the user's facial landmarks with minimal latency. Finally, the localized payment gateways COD were integrated, and the system underwent rigorous end-to-end testing before final deployment.

1.6.3 Rationale for the Selected Methodology

The rationale behind choosing this methodology is rooted in Risk Mitigation and Iterative Refinement. Because the Virtual Try-On (VTO) feature requires intensive fine-tuning for different lighting conditions and camera qualities, developing it as a separate increment allowed the team to perform targeted debugging without affecting the security of the payment or user management services. This modular approach ensures that each feature is robust and that the final system is a well-integrated, high-performance solution that meets all the requirements specified in the SRS.

Chapter 2

Problem Definition

2. Problem Definition

2.1 Problem Statement

The core problem that Lensify addresses is the lack of an interactive, accurate, and personalized experience in online eyewear shopping [10]. Traditional e-commerce platforms allow users to browse products, but they do not provide a realistic way to visualize how glasses will look on a user's face before purchase. Firstly, customers face a visualization gap, as they cannot physically try eyewear online. This limitation creates uncertainty in decision-making and reduces customer confidence, often leading to hesitation or incorrect purchases. Secondly, existing online eyewear systems offer limited personalization, where users are shown generic product listings without considering individual facial features such as face shape, size, and style compatibility[11]. This results in poor product matching and increases the likelihood of returns. Thirdly, there is a lack of intelligent recommendation and real-time integration, where users cannot simultaneously try different eyewear styles, compare options, and receive AI-based suggestions in a seamless and interactive environment. Overall, these challenges highlight the need for a smart, AI-powered solution that combines virtual try-on technology, facial analysis, and recommendation systems. Lensify aims to solve these issues by providing a real-time virtual eyewear experience with personalized product suggestions, making online eyewear shopping more accurate, engaging, and user-friendly.

2.1.1 Ineffective Virtual Product Visualization

Existing e-commerce eyewear platforms fail to provide realistic product visualization for users. Most systems only display static product images or basic filters, which do not allow customers to understand how glasses will actually appear on their face.

Without virtual try-on capability, users rely on guesswork, which leads to poor purchasing decisions, dissatisfaction, and increased return rates. This limitation directly affects user trust in online eyewear shopping systems.

2.1.2 Lack of Personalization and Intelligent Recommendations

Current eyewear platforms provide limited or no personalization based on user facial structure. Users are shown the same products regardless of whether they suit their face shape, size, or style preferences. In addition, there is no AI-based recommendation system that can analyze facial features and suggest suitable eyewear automatically. This results in a non-personalized shopping experience and reduces overall user satisfaction.

2.1.3 Low Engagement and Trust in Online Eyewear Shopping

Another major issue is the lack of engagement and trust in online eyewear platforms. Since users cannot try products physically or see realistic previews, they often hesitate before making a purchase. Furthermore, the absence of advanced visualization tools and interactive features reduces user engagement and makes the shopping experience less appealing compared to physical stores.

2.2 Deliverables and Development Requirements

The successful completion of the Lensify project is defined by the implementation of all functional and non-functional requirements, along with the following key deliverables.

2.2.1 Core Application (Lensify Platform)

The primary deliverable is a web/mobile-based AI eyewear platform that allows users to browse, try, and purchase eyewear products using virtual try-on technology. The system includes.

- User interface for product browsing
- Real-time virtual try-on module
- Face detection and tracking system
- Product catalog and filtering system

2.2.2 AI-Powered Modules

The system includes intelligent AI components that enhance user experience:

- **Virtual Try-On Engine:** Real-time overlay of eyewear on user face using camera input
- **Facial Feature Detection Module:** Detects face shape, eyes, and positioning for accurate alignment
- **Recommendation System:** Suggests suitable eyewear based on facial features and style preferences

2.2.3 User Management System

The platform includes secure user authentication and profile management:

- User registration and login
- Profile creation and management
- Saved preferences and history tracking

2.2.4 Product Management System

A structured system to manage eyewear products:

- Add/update/delete products
- Categorization of eyewear (sunglasses, prescription glasses, etc.)
- Product images and descriptions

2.2.5 Documentation and Project Artifacts

The final deliverables also include:

- Complete project report (FYP documentation)
- System design diagrams (architecture, flowcharts, UML)
- Source code and implementation files
- Requirement specification document (SRS)

2.3 Current System Analysis

Existing eyewear and e-commerce platforms only provide basic product browsing and lack interactive features. They do not support real-time virtual try-on, personalized AI recommendations, or chatbot assistance, which leads to poor user experience and uncertain purchase decisions. This highlights the need for a smart and interactive system like Lensify.

Table 2.1: Current System Analysis

Platform / System	Weaknesses	Lensify Proposed Solution
Traditional E-Commerce Sites	Only static images, no try-on feature	Real-time virtual try-on using AI
Warby Parker / Lenskart	Limited customization and region-based availability	Fully AI-based facial analysis and global accessibility
Mobile Camera Filters	Not linked with actual shopping or product catalogs	Integrated shopping + try-on + recommendation system
Manual Store Try-on	Requires physical visit and limited options	Instant online try-on with unlimited product comparison

The major gap in existing systems is the absence of a fully integrated AI-based eyewear shopping experience. Lensify bridges this gap by combining computer vision, augmented reality, and machine learning-based recommendations into a single unified platform. This integration transforms the traditional online shopping experience into a smart, interactive, and personalized system, making eyewear selection more accurate, efficient, and user-friendly.

Chapter 3

REQUIREMENT ANALYSIS

3. REQUIREMENT ANALYSIS

This chapter translates the generalized problem statement into a precise set of user goals and defined system capabilities, culminating in the complete Software Requirements Specification (SRS) for the Lensify platform. This analysis ensures that all functional and non-functional requirements are clearly identified and structured before system design and development. The requirements are derived using techniques such as use case analysis, user interaction modeling, and system behavior definition, and they remain traceable throughout the entire development lifecycle. The main focus is to ensure that the system effectively supports AI-based virtual try-on, facial analysis, recommendation services, and chatbot assistance in a unified platform.

3.1 Use Cases Diagram

The overall functionality of the Lensify AI-powered eyewear system is visually represented in the Use Case Diagram below. It defines how different users interact with the system's intelligent features such as virtual try-on, facial analysis, product recommendation, and AI chatbot assistance. The Use Case Diagram represents the functional structure of the Lensify system, showing the interaction between external actors and internal system processes. The system is designed to provide a seamless AI-based virtual eyewear experience, where users can try glasses in real-time, receive intelligent suggestions, and interact with an AI chatbot for guidance. The system acts as a bridge between computer vision, augmented reality, and machine learning-based recommendation services, ensuring a smart and interactive shopping experience.

The system boundary labeled Lensify System defines the scope of the application. All functionalities such as virtual try-on, face detection, recommendation generation, and chatbot interaction are included inside this boundary.

Anything outside the boundary (User, Admin, Camera input, etc.) interacts with the system but is not part of the internal processing logic.

3.1.1 Actors

The Lensify system involves the following primary and supporting actors.

- **User (Customer) – Primary Actor** Represents the end-user who interacts with the system to try eyewear virtually and receive AI-based recommendations.
- **AI Engine (System Core) – Processing Actor**
Represents the internal intelligent system responsible for:
 - Face detection
 - Virtual try-on processing
 - Recommendation generation
 - Chatbot response handling
- **Admin – Control Actor**
Represents the system administrator responsible for managing products and monitoring system performance.

3.1.2 Use Cases and Functional Mapping

The system is divided into multiple interconnected use cases.

A. Authentication & Access Control

- **User Login / Signup**
Users create accounts and securely access personalized features.
- **Profile Management**
Users maintain personal details, preferences, and saved eyewear history.

B. Core AI Virtual Try-On System

- **Real-Time Face Detection**
The system captures user face using a camera and detects key facial landmarks.
- **Face Shape Analysis**
The AI engine classifies face type (round, oval, square, etc.) for better product matching.
- **Virtual Eyewear Overlay**
Glasses are digitally placed on the user's face in real time using augmented reality.

- **Live Adjustment System**

Automatically adjusts glasses position when the user moves their face.

C. Product Interaction System

- **Browse Eyewear Catalog**

Users can explore different categories (sunglasses, prescription glasses, etc.).

- **Product Filtering**

Users filter products based on style, shape, colour, and price.

- **Product Comparison**

Users compare multiple eyewear options side-by-side using virtual preview.

- **Wishlist / Save Product**

Users save preferred eyewear for future selection.

D. AI Recommendation System

- **Facial Feature-Based Recommendation**

AI analyzes facial structure and suggests suitable eyewear.

- **Style Preference Learning**

System adapts suggestions based on user interaction history.

- **Smart Ranking of Products**

Best-suited eyewear is prioritized in results.

E. AI Chatbot Assistant

- **User Query Handling**

Chatbot answers questions like Which glasses suit me best?

- **Style Guidance**

Suggests eyewear based on fashion trends and face shape.

- **System Navigation Help**

Helps users understand how to use virtual try-on features.

- **Product Assistance**

Provides quick recommendations and product information.

F.Admin Panel Functionality

- **Product Management**

Admin can add, update, or remove eyewear products.

- **User Monitoring**

Admin monitors user activity and system usage trends.

- **System Performance Monitoring**

Tracks AI accuracy, response time, and system efficiency.

3.1.3 AI System Internal Processing Flow

The AI Engine performs the following internal operations.

- Captures user face via camera input
- Detects facial landmarks (eyes, nose, face shape)
- Processes data using ML model
- Matches face type with eyewear database
- Generates recommendation score
- Sends output to chatbot for user interaction support

3.1.4 Summary of Interactions

The Lensify system operates as a multi-layer AI architecture.

- The User layer interacts with virtual try-on, browsing, and chatbot features.
- The AI Engine layer processes facial recognition, AR rendering, and recommendation logic.
- The Admin layer ensures system stability, product control, and data management.
- This structure ensures a fully integrated AI-powered eyewear ecosystem that eliminates guesswork and provides a personalized, real-time shopping experience.

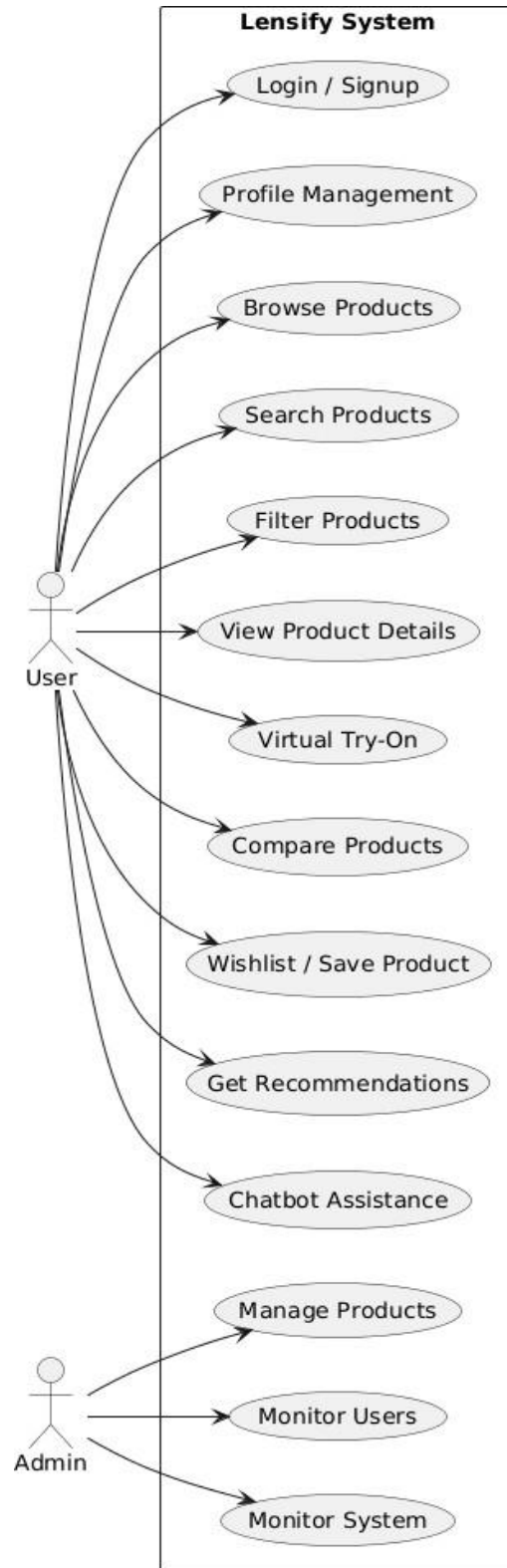


Figure 3.1 Use Case Diagram of Lensify

The Use Case Diagram in Figure 3.1 organizes the system requirements of the Lensify platform into logical functional blocks.

The User (Customer) interacts with the system to browse products, apply filters, and view details. The main feature is the Virtual Try-On, which allows users to see eyewear in real time. Users can also compare products, save items, get AI-based recommendations, and use the chatbot assistant for guidance and support.

The admin acts as the system controller, responsible for managing products, monitoring users, and ensuring system performance.

3.2 Detailed Use Case

A detailed description of key use cases in the Lensify system ensures a clear understanding of user interactions, system responsibilities, and expected functionality. This section expands the main features into step-by-step processes to define how the system behaves during real usage.

The focus is on critical operations such as Virtual Try-On, product browsing, AI-based recommendations, and chatbot interaction. For example, in the virtual try-on process, the user activates the camera, the system captures facial data, and eyewear is displayed in real time. Similarly, in the recommendation process, the system analyzes user preferences and suggests suitable products.

These detailed scenarios help in mapping user actions to system responses, ensuring that all functional requirements are properly implemented. They also provide a strong foundation for system design, development, and testing, making sure the platform delivers a smooth, interactive, and personalized eyewear experience.

3.2.1 Detailed Use Case: Secure User Registration

This case outlines the process for a new user to securely establish an account, defining their primary role on the platform and ensuring the integrity of the user base.

Table 3.1: Secure User Registration (Sign Up)

Attribute	Description
Use Case ID	FR-01
Use Case Name	User Registration
Actors	User
Description	Allows a new user to create an account to access Lensify features.
Trigger	User opens the system and selects Sign Up”.
Pre-Conditions	1. User has internet access. 2. User opens the application.
Normal Flow	1. User navigates to registration page. 2. User enters name, email, and password. 3. System validates input. 4. System stores user data. 5. User account is created successfully.
Alternative Flows	A1: Missing Data → System prompts user to fill required fields.
Exceptions	E1: Duplicate Email → System shows error and asks user to login.
Post Conditions	1. User account is created. 2. User can proceed to login.

3.2.2 Detailed Use Case: Secure User Authentication

This use case manages the process by which a registered user securely accesses the platform's personalized features, applying role-based access to the correct dashboard.

Table 3.2: Detailed Use Case: Secure User Authentication (Login)

Attribute	Description
Use Case ID	FR-02
Use Case Name	Login
Actors	User, Admin
Description	Allows registered users to securely access the system.
Trigger	User selects Login.
Pre-Conditions	1. User must have a registered account.
Normal Flow	1. User enters email and password. 2. System verifies credentials. 3. Access is granted.
Alternative Flows	A1: Forgot Password → Reset option provided.
Exceptions	E1: Invalid credentials → Access denied.
Post Conditions	User is successfully logged into the system.

3.2.3 Detailed Use Case: Virtual Try-On

This is the central feature where the user interacts with eyewear products in real time using the camera-based virtual try-on system. It allows users to visualize how different glasses will look on their face before making a purchase decision, improving accuracy and user confidence in online eyewear selection.

Table 3.3: Detailed Use Case: Virtual Try-On

Attribute	Description
Use Case ID	FR-03 (Virtual Try-On)
Use Case Name	Virtual Eyewear Try-On
Actors	User
Description	Allows the user to try different eyewear virtually in real time using the camera, enabling visualization before purchase.
Trigger	User selects the “Try On” option on a product.
Pre-conditions	1. User must be logged in. 2. Camera permission must be enabled.
Normal Flow	1. User opens the virtual try-on feature. 2. System activates camera and detects user face. 3. User selects an eyewear product. 4. System overlays selected glasses on user’s face in real time. 5. System adjusts eyewear position according to face movement.
Alternative Flows	A1: Camera permission not granted → System requests permission again. A2: Face not detected → System asks user to adjust position.
Exceptions	E1: Camera error → System shows error and stops try-on process.
Post Conditions	1. User successfully views eyewear on face in real time. 2. User can proceed to save, compare, or explore other products.

3.2.4 Detailed Use Case: Browse Products

This case describes the process where a user explores the eyewear catalog, views available products, and accesses detailed information to support selection and decision-making before using features like virtual try-on or product saving.

Table 3.4: Detailed Use Case: Browse Products

Attribute	Description
Use Case ID	FR-04
Use Case Name	Browse Eyewear Products
Actors	User
Description	Allows users to explore available eyewear products in the system, view categories, and access detailed information about each product to support selection and decision-making.
Trigger	User opens the product section or homepage of the application.
Pre-Conditions	1. System must be running. 2. Product data must be available in the database.
Normal Flow	1. User navigates to the product section. 2. System retrieves product data from the database. 3. System displays a list/grid of eyewear products (e.g., sunglasses, prescription glasses). 4. User scrolls and browses through available products. 5. User selects a product. 6. System displays detailed product information (image, name, price, description, and features).
Alternative Flows	A1: No Products Available → System displays a “No products available” message. A2: Slow Loading → System shows loading indicator until data is retrieved.
Exceptions	E1: Data Fetch Error → System displays error message and provides retry option.
Post Conditions	1. Products are successfully displayed to the user. 2. User can proceed to actions like virtual try-on, filtering, comparison, or saving products.

3.2.5 Detailed Use Case: AI Recommendations

This case describes the process where the system provides personalized eyewear suggestions to the user based on their interactions, preferences, and virtual try-on history to support better decision-making before purchase.

Table 3.5: Detailed Use Case: AI Recommendations

Attribute	Description
Use Case ID	FR-05
Use Case Name	AI-Based Eyewear Recommendations
Actors	User
Description	Provides personalized eyewear suggestions to users based on their preferences, browsing behavior, and interaction with the virtual try-on system.
Trigger	User requests recommendations or opens the recommendation section.
Pre-Conditions	1. User must be logged in. 2. System must have user interaction data (browsing or try-on history).
Normal Flow	1. System collects user interaction data (browsed products, try-on results, saved items). 2. AI recommendation system analyzes user preferences. 3. System matches user profile with suitable eyewear products. 4. Recommended products are displayed in ranked order.
Alternative Flows	A1: No user data available → System shows general popular or trending eyewear recommendations.
Exceptions	E1: Recommendation engine failure → System displays retry option or default product list.
Post Conditions	1. Personalized eyewear suggestions are shown to the user. 2. Recommendations are updated for future interactions.

3.2.6 Detailed Use Case: AI Chatbot Assistance

This case describes the process where the system provides real-time assistance to users through an AI chatbot, helping them with eyewear selection, system navigation, and general support within the Lensify platform.

Table 3.6: Detailed Use Case: AI Chatbot Assistance

Attribute	Description
Use Case ID	FR-06
Use Case Name	AI Chatbot Assistance
Actors	User
Description	Provides real-time guidance, support, and product-related assistance to users during their interaction with the Lensify system.
Trigger	User opens the chatbot or sends a message.
Pre-Conditions	1. System must be active and running. 2. User must have access to the application interface.
Normal Flow	1. User opens chatbot interface. 2. User enters a query related to eyewear, recommendations, or system usage. 3. System processes the query using AI chatbot module. 4. Chatbot generates a relevant response. 5. Response is displayed to the user in real time.
Alternative Flows	A1: Query not understood → System provides general help or suggests rephrasing.
Exceptions	E1: Chatbot service failure → System displays error message and retry option.
Post Conditions	1. User receives appropriate guidance or information. 2. Chat interaction is stored for future improvement of responses.

3.3 Functional Requirements

Functional requirements define the core features and services that the Lensify system must provide to the user. These requirements are directly implemented in the system to ensure full functionality of the AI-powered eyewear platform.

3.3.1 User Registration and Authentication

The Lensify platform provides a secure and user-friendly registration and authentication system to ensure safe access for all users, including customers and administrators.

The system allows users to create an account by selecting their role (such as Customer or Admin) and entering required details like name, email, and password. To maintain security and prevent fake accounts, email verification is mandatory before activating any account.

During registration, the system enforces strong password rules to enhance security. Users are required to create passwords with a minimum length of 8 characters, including at least one numeric value and one special character. This helps protect user accounts from unauthorized access.

3.3.2 Product Browsing and Interaction

The system shall allow users to explore and interact with available eyewear products in a structured and user-friendly way.

- The system shall display a list of available eyewear products with images and basic details.
- The system shall allow users to search products by name or category (e.g., sunglasses, prescription glasses).
- The system shall provide filtering options based on style, price range, and category.
- The system shall allow users to view detailed product information including description, price, and features

3.3.3 Virtual Try-On System

The system shall provide a real-time virtual try-on feature that allows users to visualize eyewear on their face before purchase.

- The system shall allow users to activate camera-based face detection for real-time processing.

- The system shall display selected eyewear on the user's face using augmented reality overlay.
- The system shall continuously adjust the eyewear position according to face movement and orientation.
- The system shall allow users to try multiple eyewear products for comparison before making a final decision.

3.3.4 AI Recommendations

The system shall provide personalized eyewear recommendations to improve user selection and shopping experience.

- The system shall analyse user browsing history, saved products, and virtual try-on interactions.
- The system shall suggest eyewear based on user preferences, style behaviour, and face structure analysis.
- The system shall rank products according to relevance, suitability, and user interest.
- The system shall continuously update recommendations based on new user activity.

3.3.5 AI Chatbot Assistance

The system shall include an AI chatbot to provide real-time user support and guidance.

- The chatbot shall answer questions related to eyewear products and features.
- The chatbot shall guide users on how to use virtual try-on and other system functionalities.
- The chatbot shall provide style suggestions based on user queries and preferences.
- The chatbot shall assist users in navigating the Lensify platform.

3.3.6 Product Management

The system shall allow the administrator to manage eyewear product data.

- The admin shall be able to add new eyewear products with complete details.
- The admin shall update or delete existing products when required.
- The system shall ensure that all product changes are reflected instantly to users.

3.3.7 User Profile Management

The system shall allow users to manage their personal profile and preferences.

- The system shall allow users to update personal information such as name, email, and password.
- The system shall store user preferences for eyewear style and category.
- The system shall maintain user history including viewed products and try-on records.

3.3.8 Product Search and Filtering Enhancement

The system shall provide advanced search and filtering options for better product discovery.

- The system shall allow users to search eyewear by name or keyword.
- The system shall support filters such as price range, style, and category.
- The system shall allow users to quickly refine results for better selection.

3.3.9 Wishlist Management

The system shall allow users to save or bookmark preferred eyewear products.

- The system shall allow users to add products to a wishlist.
- The system shall allow users to view saved items anytime.
- The system shall allow users to remove items from wishlist when needed.

3.3.10 Payment Integration

The system shall support order placement through the Cash on Delivery (COD) payment method

- The system shall allow users to place orders using Cash on Delivery (COD).
- The system shall generate an order confirmation after successful checkout.
- The system shall store order and delivery details in the database.
- The system shall allow users to track the status of their orders.

3.4 Non-Functional Requirements

Non-Functional Requirements define the quality attributes, constraints, and performance standards of the Lensify system. These requirements ensure that the system is fast, secure, scalable, and easy to use.

They focus on how well the system performs its functions, including response time, reliability, data security, and user experience. Lensify must support real-time virtual try-on and AI features without delay, protect user data through secure authentication, and provide a simple, user-friendly interface. The system should also be scalable for future growth and easy to maintain for updates and improvements.

3.4.1 Performance

The system must provide fast response time and smooth user interaction, especially for real-time features such as virtual try-on and product browsing.

- The system shall load product listings within 2–3 seconds under normal network conditions.
- The virtual try-on feature shall provide real-time face tracking with minimal delay.
- AI-based recommendations and chatbot responses shall be processed without noticeable delay for the user.
- Heavy processing tasks (e.g., AI analysis) shall be optimized to avoid system lag during user interaction.

3.4.2 Scalability

The system shall be designed to handle increasing users and data without performance degradation.

- The architecture shall support future expansion of users, products, and AI features.
- The system shall allow independent scaling of major modules such as virtual try-on, recommendation system, and chatbot.
- The system shall support growing product database and user interaction history efficiently.

3.4.3 Security and Privacy

The system shall ensure protection of user data and secure access to all features.

- The system shall use secure authentication for user login and registration.
- User passwords shall be stored in encrypted form.
- All communication between client and server shall be protected using secure protocols (HTTPS/SSL).
- The system shall implement role-based access control to separate user and admin functionalities.
- User data (profile, preferences, and history) shall be kept private and protected from unauthorized access.

3.4.4 Usability

The system shall provide an easy-to-use and user-friendly interface for all types of users.

- The system shall have a simple and intuitive UI for smooth navigation.
- The application shall be fully responsive and work on mobile, tablet, and desktop devices.
- AI features such as recommendations and chatbot shall be easy to understand and use.
- The virtual try-on feature shall be accessible with minimal steps for the user.

3.4.5 Maintainability

The system shall be designed for easy maintenance, updates, and future enhancements.

- The system shall follow modular architecture to separate different functionalities (Try-On, AI, Chatbot, Product Management)
- The codebase shall be well-structured and documented for future developers.
- New features such as new eyewear categories or AI improvements can be added without affecting existing system modules.
- The system shall support continuous improvement and updates with minimal downtime.

Chapter 4

Design and Architecture

4. Design and Architecture

4.1 System Architecture

The Lensify system is designed using a Layered (Multi-Layered / N-Tier Architecture.) Architecture, which consists of the Presentation Layer, Application Layer (Business Logic), and Data Layer. This architecture ensures modularity, scalability, and efficient integration of AI-based components such as Virtual Try-On and Recommendation Engine.

The system combines modern web technologies with AI modules to provide a seamless and intelligent eyewear shopping experience. Each layer performs a specific role and communicates through secure APIs, enabling smooth data flow and maintainability.

4.2 Architecture Design Diagram

The architecture of Lensify follows an N-tier structure, dividing the system into logical layers: Client Layer, Presentation Layer, Processing Layer (AI & VTO), Application Layer, and Data Layer. This separation ensures better performance, scalability, and clear responsibility of each component.

Detailed Layer Breakdown:

1. Client Tier (Frontend):

- The Client Tier acts as the user interaction point.
- It consists of a web-based interface built using React.js.
- Users can:
 - Browse products
 - Use Virtual Try-On
 - View recommendations
 - Manage cart and orders
 - The interface is fully responsive, supporting desktop and mobile browsers.

2. Presentation Layer (UI & API Handling)

- This layer manages communication between frontend and backend.
- It handles:
 - User requests (HTTP/REST APIs)
 - Data validation
 - UI rendering
 - Supports real-time features like:
 - Virtual Try-On activation
 - Dynamic product updates
- Ensures smooth user experience with responsive design and accessibility.

3. Virtual Try-On & AI Processing Layer

- This is a core intelligent layer of the system.
- **Technologies used:**
 - MediaPipe Face Mesh (face detection)
 - TensorFlow.js (AI processing)
- **Functions:**
 - Detect facial landmarks
 - Identify face shape
 - Overlay glasses in real-time
 - Generate AI-based recommendations

4. Application Layer (Business Logic - Backend)

- Implemented using Node.js and Express.js.
- Handles all core functionalities:
 - User authentication

- Product management
- Cart and order processing
- Prescription handling
- **Integrates with:**
 - AI services
 - Payment gateways (Cash on Delivery (COD))
- Ensures secure and efficient processing of user actions.

5. Payment Processing Layer

- Manages all financial transactions.
- Integrated with:
 - Cash on Delivery (COD)
- Responsibilities:
 - Secure payment processing
 - Transaction validation
 - Order confirmation updates
- Uses encryption to ensure data security.

6. Data Layer (Database - MongoDB Atlas)

- Uses MongoDB Atlas (NoSQL cloud database).
- Stores:
 - User data
 - Products
 - Prescriptions

- Orders
- Payments
- Chat logs
- Features:
 - High availability
 - Fast data retrieval using indexing
 - Scalable cloud storage

7. Admin Dashboard Module

- Provides system control to administrators.
- Features:
 - Product management (Add, Update, Delete)
 - Order tracking
 - User management
 - Analytics monitoring

Key Architectural Highlights

Modularity: Each system component (AI, VTO, Payment, Backend) works independently.

Scalability: The layered design allows easy scaling of AI modules and backend services.

Real-Time Processing: Virtual Try-On works instantly using AI and browser-based processing.

Security: Secure authentication, encrypted payments, and protected APIs.

Performance Optimization: Heavy AI tasks are handled efficiently without slowing down the system.

Maintainability: Clean separation of layers makes future updates and improvements easier.

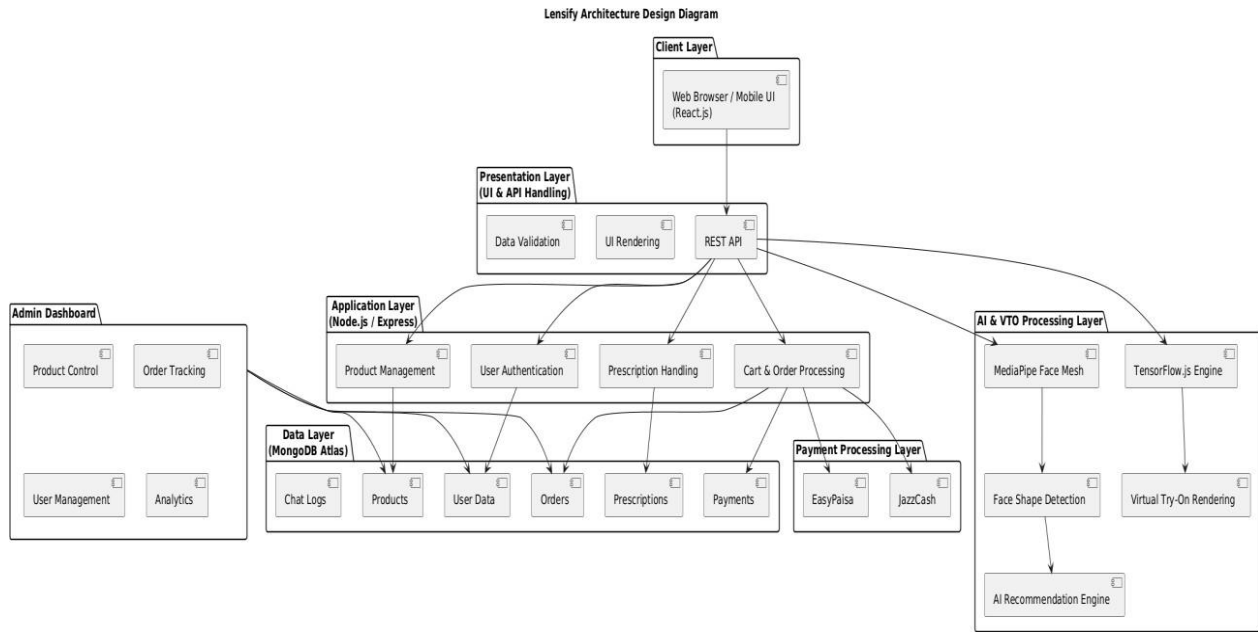


Figure 4.1 Architecture Design Diagram of Lensify

4.2.1 Layer Components and Responsibilities

Presentation Layer: Acts as the client-facing interface of the system. It is developed using React.js to provide responsive and user-friendly design. Its main function is to handle user inputs, display products, recommendations, Virtual Try-On results, and order information fetched from the backend, while also ensuring smooth communication with the Application Layer through REST APIs.

Application Layer: Contains the core business logic of the Lensify system and is implemented using Node.js and Express.js. It manages all main system operations including user authentication, product management, cart and order processing, and prescription handling. It also integrates AI services for recommendations and face-shape detection, and connects with external payment gateways to complete transactions securely.

AI Processing Layer: This layer handles all intelligent and real-time processing features of the system. It uses MediaPipe Face Mesh for detecting facial landmarks and TensorFlow.js for rendering eyewear on the user's face in real time. It is also responsible for generating face-shape detection results and providing AI-based personalized eyewear recommendations.

Payment Layer: This layer manages order processing within the system. It supports Cash on Delivery (COD) as the payment method instead of online payment gateways. It handles order confirmation, updates order status, and ensures proper tracking of customer orders after checkout.

Data Layer: The Data Layer is implemented using MongoDB Atlas (NoSQL database). It stores all system data including user information, product details, prescriptions, orders, payments, and chatbot logs. It ensures fast data retrieval, scalability, and high availability for the system.

Admin Layer: This layer provides administrative control over the system. It allows admins to manage products, monitor orders, handle user accounts, and view system analytics to ensure smooth system operations.

4.3 Data Representation

The Data Design of the Lensify system focuses on defining the core structural components of the application through a Class Diagram, which acts as the main blueprint for both relational and non-relational data organization. It clearly defines how different entities such as users, products, prescriptions, orders, and payments are structured and how they interact with each other within the system.

This design ensures data consistency, integrity, and scalability across all modules, including Virtual Try-On, AI-based recommendations, cart management, and payment processing. By properly defining classes and their relationships, the system maintains a well-organized data flow that supports efficient storage, retrieval, and processing of information in MongoDB Atlas.

4.3.1 Class Diagram

The Class Diagram of the Lensify system represents the static structure of the application, showing how different entities interact to support an AI-powered eyewear e-commerce platform. The system is designed using an object-oriented approach where all core components are organized into classes to ensure reusability, scalability, and maintainability.

1. User Hierarchy (Inheritance)

The system is based on a common **User superclass**, which is inherited by different user roles to manage access and functionality efficiently.

User (Parent Class):

This is the base class for all system users. It contains common attributes such as userID, name, email, password, phone, address, and role, along with general methods like login(), updateProfile(), and logout().

- **Customer (Child Class):** Inherits from User and represents the end customer of the Lensify platform. It is responsible for browsing products, using Virtual Try-On, placing orders, and managing prescriptions. Key methods include browseProducts(), useVirtualTryOn(), addToCart(), and placeOrder().
- **Admin (Child Class):** Inherits from User and is responsible for system management. It handles product management, user control, and order monitoring. Key methods include manageProducts(), manageOrders(), and manageUsers().

2. Core Entities

These classes represent the main business objects of the Lensify system.

- **Product:** Represents eyewear items available in the system. It includes attributes such as productID, name, brand, price, category, frameShape, lensType, stock, and image. It supports operations like viewDetails().
- **Cart:** Represents the shopping cart of a user. It contains attributes like cartID, userID, items, and totalAmount. It supports operations such as addItem(), removeItem(), and calculateTotal().
- **CartItem:** Represents individual products inside the cart. It includes productID, quantity, and selectedLensType.
- **Prescription:** Stores user eye prescription details such as prescriptionID, leftEye, rightEye, PD, and additional notes.
- **Order:** Represents a confirmed purchase. It includes orderID, userID, items, totalAmount, status, shippingAddress, and orderDate. It manages order lifecycle from placement to delivery.
- **Payment:** Handles financial transactions with attributes such as paymentID, orderID, amount, paymentMethod, status, and transactionID.

3. AI & Virtual Try-On Module

AIEngine: This class provides intelligent features to enhance user experience. It is responsible for:

- faceShapeDetection() using MediaPipe Face Mesh
- virtualTryOn() using TensorFlow.js
- recommendProducts() based on user behavior and face shape
- analyzePreferences() for personalization

4. Enumerations (Enums)

These define fixed values used throughout the system to maintain consistency.

- **UserRoleEnum:** CUSTOMER, ADMIN
- **OrderStatusEnum:** PENDING, PAID, PROCESSING, SHIPPED, DELIVERED, CANCELLED
- **ProductCategoryEnum:** GLASSES, SUNGLASSES, CONTACT_LENSES

Summary of Relationships

- A Customer can have multiple Orders and one Cart (1-to-many and 1-to-1 relationship).
- A Cart contains multiple CartItems.
- An Order contains multiple Products through CartItems.
- A Customer can have multiple Prescriptions linked to Orders.
- A Payment is linked to one Order only.
- The AIEngine interacts with Product, User, and Prescription classes to provide intelligent recommendations and Virtual Try-On results.

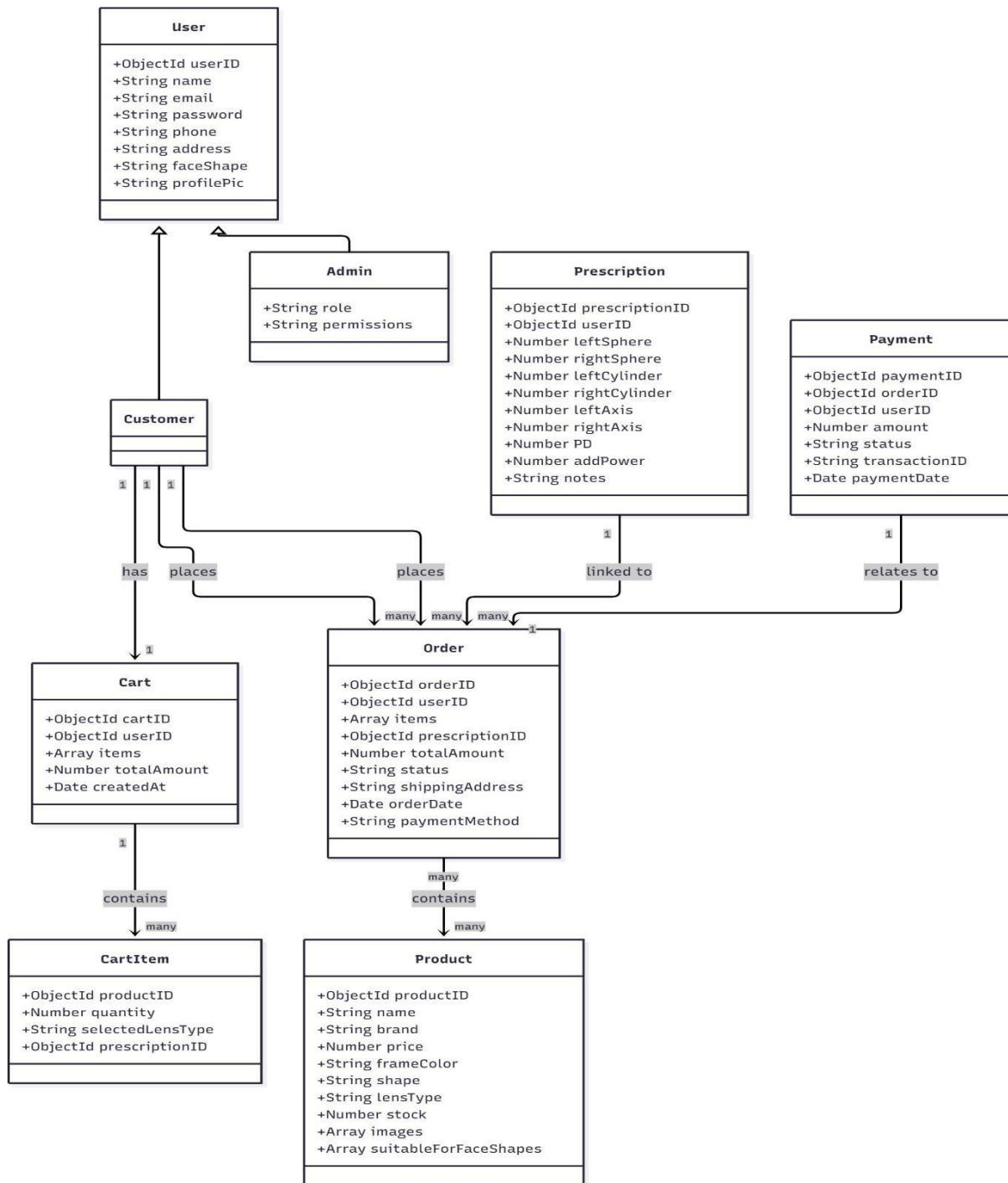


Figure 4.2 Class Diagram of Lensify

The **Class Diagram** provides a static object-oriented structural view of the Lensify system, defining its main entities, attributes, and relationships that support the AI-powered eyewear e-commerce platform.

Core Entities and Inheritance: The system is built around a base class **User**, which contains common attributes such as *userID*, name, email, password, phone, address, role, and *profileImage*. This class is extended into specialized roles including **Customer** (who browses products, uses Virtual Try-On, and places orders) and **Admin** (who manages products, users, and system operations). This inheritance structure ensures reusability and role-based access control across the system.

Product: Represents eyewear items such as glasses and lenses, including attributes like name, brand, category, price, stock, and frameShape compatibility.

Cart & CartItem: Manage user-selected products before checkout, including quantity, selected lens type, and optional prescription linkage.

Order & OrderItem: Represent confirmed purchases, storing order details, selected products, prescription data, total amount, shipping address, status, and payment method.

Prescription: Stores optical details such as sphere, cylinder, axis, PD, and add power.

Payment: Handles financial transactions including amount, transaction ID, status, and payment date, linked directly to orders.

AI and Intelligent Processing Support (Embedded Functionality):

Instead of being a separate structural entity, AI functionality is integrated into system behavior. It supports.

- Face shape detection for product suitability
- Recommendation logic based on user interaction and product attributes
- Smart filtering for personalized shopping experience
- These AI-driven processes enhance decision-making during product browsing, Virtual Try-On, and recommendation generation.

Relationships and System Flow:

The relationships define how objects interact within the system:

- A Customer can have one cart and multiple orders.
- A Cart contains multiple CartItems, each linked to a Product.
- An Order consists of multiple OrderItems and is associated with one Payment record.

- A Prescription can be linked to cart items or orders for customized lens processing.
- The Admin manages product inventory and system-level data.

4.3.2 Data Dictionary Summary

The Class Diagram represents a high-level data dictionary of the Lensify system, defining key entities such as User, Product, Cart, Order, Prescription, and Payment. Each entity includes essential attributes for system operations, including user details, product information, shopping data, prescription values, and transaction records, ensuring consistent and structured data management across the platform.

Table 4.1:Data Dictionary Summary

Entity	Key Field	Data Type	Constraint/Purpose
User	role	roleEnum	Defines access level: Admin, Customer
Customer	faceShape	String	Used for Virtual Try-On and recommendations
Product	frameShape	String	Used for AI-based product matching
Product	price	float	Used in filtering and recommendation ranking
CartItem	quantity	Integer	Defines number of selected items
Order	status	orderStatusEnum	Tracks order lifecycle (Placed, Paid, Shipped, Delivered)
Prescription	PD	float	Used for lens customization accuracy
Payment	transactionID	String	Unique identifier for payment verification
Review	rating	float	Numerical feedback for product evaluation
Review	sentiment_score	sentimentEnum	AI-based sentiment classification (Positive, Neutral, Negative)

4.4 Process Flow/Data Flow Representation

The Process Flow Diagram (PFD) of **Lensify** illustrates the complete user journey and system-level data movement across key modules such as authentication, product browsing, Virtual Try-On, AI recommendations, cart management, order processing, and payment handling. It also represents how data flows between frontend, backend services, AI modules, and the database to deliver a seamless e-commerce experience.

4.4.1 Process Flow Diagram

The process begins with user authentication, leading to role-based access to the system, followed by product browsing, Virtual Try-On, AI-based recommendations, cart and order processing, and payment completion. This flowchart represents the complete operational workflow of the Lensify platform, mapping the user journey from initial login to final order confirmation. It illustrates the logical flow of user actions, system responses, decision points, and interactions between customers, admin, AI modules, and the database within the system.

Detailed Workflow Phases:

1. Authentication Phase:

The process starts when the user logs in or registers on the system. The system verifies credentials and identifies the user role (Customer or Admin). If authentication fails, an error message is displayed and the user is redirected to retry login. If successful, the user is redirected to the main dashboard.

2. Product Browsing & Interaction Phase:

After login, the customer accesses the product catalog. The system retrieves product data from the database and allows users to search, filter, and view detailed eyewear information. This stage forms the base input for AI-driven features.

3. Virtual Try-On & AI Recommendation Phase:

The user activates the Virtual Try-On feature using webcam or uploaded image. The system performs face detection using AI models and overlays selected eyewear in real time. Based on

detected face shape and user behaviour, the AI Recommendation Engine suggests suitable eyewear products.

4. Cart & Order Processing Phase:

The user selects desired products and adds them to the cart. The system calculates the total price dynamically and allows users to proceed to checkout. At this stage, prescription details (if required) are also linked with the order.

5. Payment Processing Phase:

The user the Cash on Delivery (COD) payment option and confirms the order. The system records the order details, updates the order status as Pending Payment/Processing, and generates an order confirmation. Payment is collected from the customer upon successful delivery of the order.

6. Order Completion & Data Update Phase:

Once payment is successful, the order is stored in the database with updated status. The system updates inventory, stores transaction logs, and provides order tracking details to the user.

7. Admin Monitoring Phase:

The admin can monitor products, orders, and user activity through the dashboard. This ensures system control, data consistency, and operational management.

8. System Output Phase:

Finally, the system provides outputs such as order confirmation, AI recommendations, Virtual Try-On results, and order tracking updates, completing the full workflow cycle.

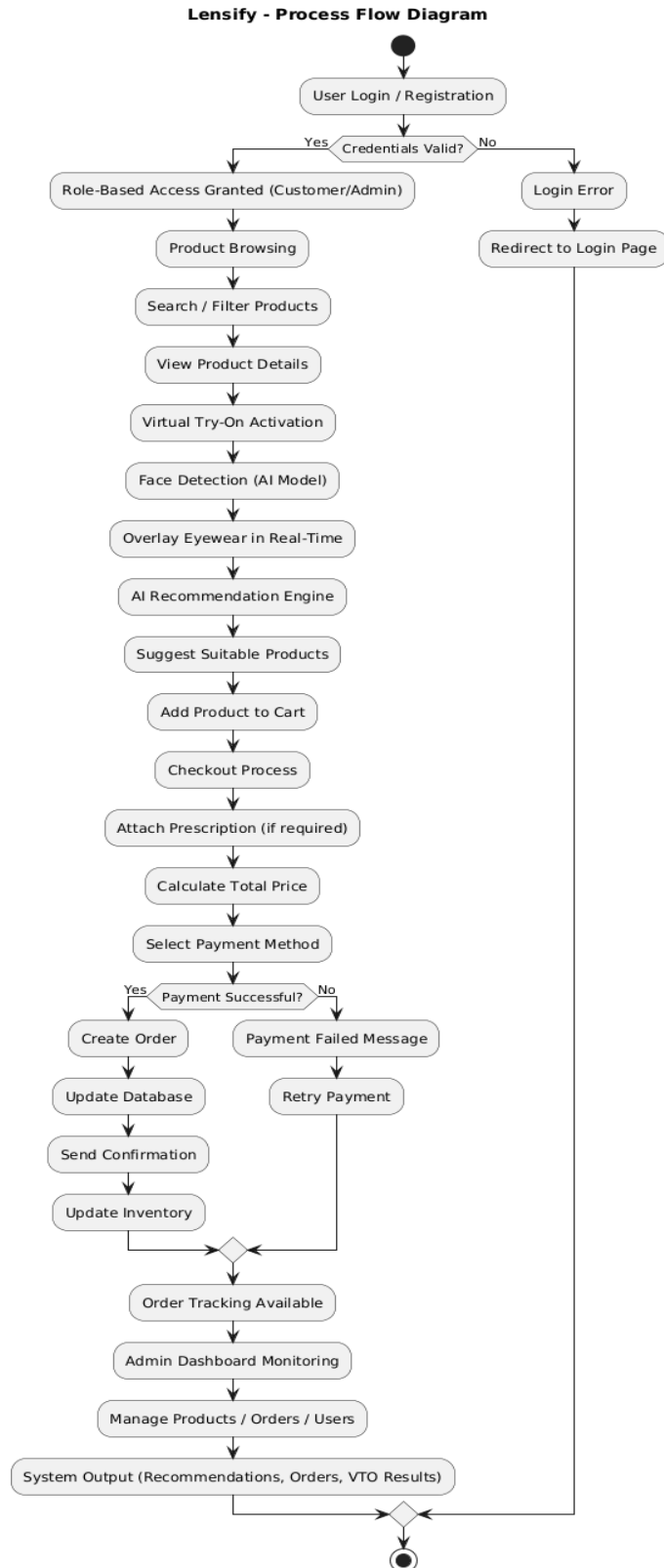


Figure 4.3 Process Flow Diagram of Lensify

The **Process Flow Diagram** is a visual representation of the core operational steps within the Lensify system, showing the movement of control and data through the platform's major functionalities, from user access to final order completion and system reporting.

1. Authentication and Access: The process begins with User Authentication, where the user logs in or registers using credentials. The system verifies identity and determines the user role (Customer/Admin). If authentication is successful, access is granted and the user is redirected to the main dashboard; otherwise, an error message is displayed and the user is asked to retry login.

2. Product Interaction and Order Processing: Once access is granted, the user enters the core system flow starting with Product Browsing, where eyewear products are viewed, searched, and filtered. The user may then activate the Virtual Try-On feature, where AI detects facial landmarks and displays real-time frame overlay. Based on user interaction, the AI Recommendation System suggests suitable eyewear.

After selection, products are added to the Cart, and the system calculates total cost. The user proceeds to checkout, optionally enters prescription details, and confirms the order. The system then processes payment through integrated gateways Cash on Delivery (COD) and updates order status upon successful transaction.

3. Order Completion and System Management: After successful payment, the system creates an Order Record, updates inventory, and stores transaction data in the database. The user can track order status until delivery completion. Meanwhile, the Admin Dashboard allows monitoring of users, products, orders, and system activity to ensure smooth platform operation.

4. System Output Phase: The process concludes with system-generated outputs such as order confirmation, AI-based recommendations, Virtual Try-On results, and order tracking updates, completing the full e-commerce lifecycle of the Lensify platform.

4.5 Design Models

The system includes UML design models to represent the dynamic behavior, interactions, and state management of the Lensify platform. These models include class, sequence, and state diagrams to clearly define system structure and process flow.

4.5.1 Order State Diagram

The Order State Diagram represents the complete lifecycle of an order in the Lensify system, managed through a Finite State Machine (FSM). It ensures controlled transitions between states such as order placement, payment verification, processing, shipping, delivery, and cancellation. This mechanism maintains transactional integrity, secure payment handling, and proper order tracking throughout the system.

Detailed Workflow Steps:

1. Cart State (Initial State)

The process begins when a customer adds eyewear products to the cart after browsing and optionally selecting prescription details. At this stage, the order is not yet confirmed and can still be modified or removed.

2. Order Placement (Placed State)

Once the customer proceeds to checkout and confirms the purchase, the system generates an order record. The order status is updated to Placed, and all selected products, prescription details, and shipping information are stored in the database.

3. Payment Processing (Cash on Delivery State)

The user confirms the order using the Cash on Delivery (COD) payment method.

- The system records the order details and delivery information.
- The order status changes to Processing after successful order confirmation.
- The system allows administrators to manage and update the order status during fulfillment.
- This step ensures accurate order processing and delivery management.

4. Order Processing State

After successful payment, the order enters the Processing state. In this phase, the admin verifies product availability, prescription accuracy, and prepares the order for fulfillment. Inventory is also updated accordingly.

5. Shipping State

Once verification is completed, the order is marked as Shipped and handed over to the delivery service. Tracking information is generated and made available to the customer.

6. Delivered State (Final State)

When the customer receives the product, the order is marked as Delivered. At this stage, the transaction is considered complete, and the system may allow the user to leave a review or feedback.

7. Cancelled / Returned State (Alternative Flow)

If the customer cancels the order before shipping, or requests a return after delivery (based on policy), the order transitions to the Cancelled/Returned state. This ensures proper handling of exceptions and post-purchase issues.

Key Technical Features:

- Strict FSM-based order lifecycle control
- Secure payment verification before order confirmation
- Real-time status updates for tracking
- Admin-controlled processing before shipping
- Support for cancellation and return handling
- Ensures data consistency across order and payment modules

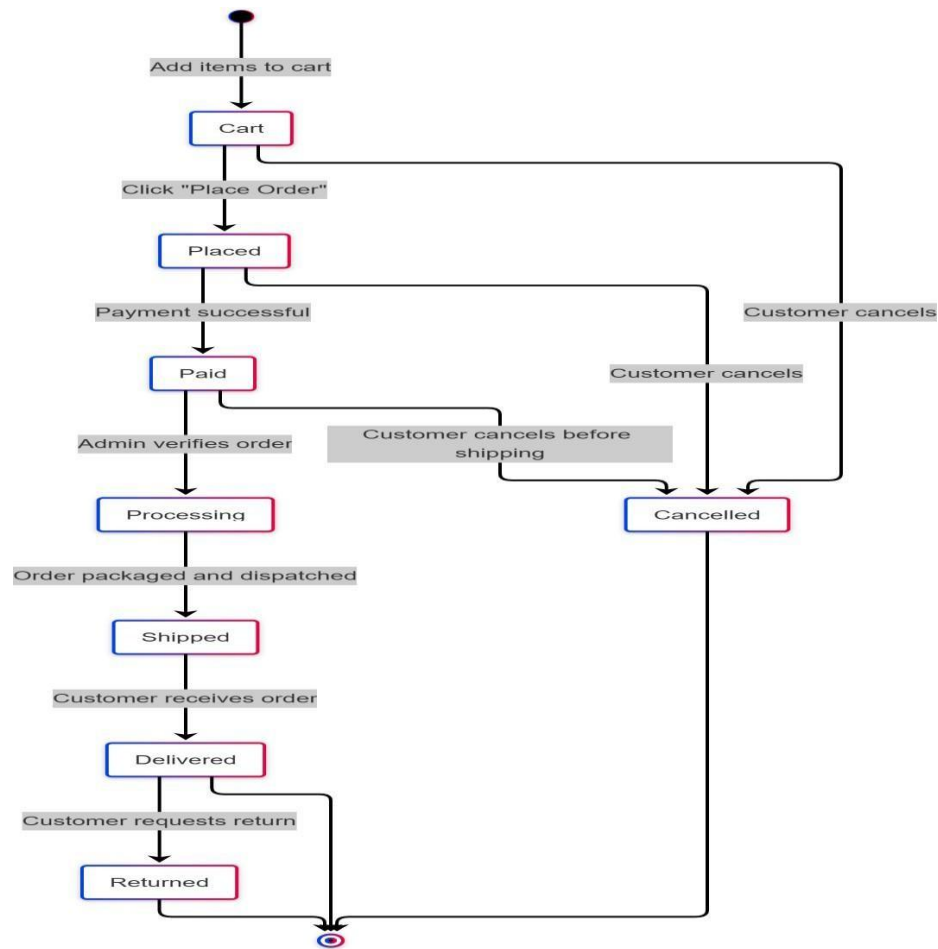


Figure 4.4 State Transition Diagram Order Lifecycle

The **State Diagram for the Order** (Product Purchase) entity in the Lensify system defines the complete lifecycle of an eyewear order as a Finite State Machine (FSM). It ensures proper control over order processing, payment validation, and delivery workflow, maintaining system consistency and reliability.

The diagram represents the mandatory sequence of states an order passes through:

- **Cart → Placed:** The order begins when a user adds products to the cart and proceeds to checkout. Once the user confirms the order, it moves to the *Placed* state.
- **Placed → Paid:** After successful payment through integrated gateways (Cash on Delivery), the order status is updated to *Paid*.

- **Paid → Processing:** The system verifies payment and order details, then prepares the eyewear (including prescription validation and inventory check).
- **Processing → Shipped:** Once packaging is completed, the order is handed over to the delivery service and marked as *Shipped*.
- **Shipped → Delivered:** The order reaches the customer and is marked as successfully *Delivered*.
- **Delivered → Returned (optional):** If the customer raises a return request (based on policy), the order transitions to *Returned*.
- **Cancellation Handling:** An order can be cancelled only before or during early processing stages depending on system rules, ensuring fairness and preventing invalid transactions. This state model ensures that critical operations such as payment confirmation, order fulfillment, and inventory updates occur in a controlled sequence. It prevents invalid transitions and guarantees secure and reliable order processing within the Lensify platform.

4.5.2 Client / Order Processing Sequence Diagram

The Sequence Diagram illustrates the dynamic flow of interactions between the user and system components during the core process of product purchase, including authentication, product selection, and order processing in the Lensify platform. It represents the chronological exchange of messages between the Client, Web Application (Frontend), Backend Services, AI modules (if required), Database, and Payment Gateway.

1. **Authentication Flow:** The process begins when the Client logs into the system through the Web Application. The UI sends a Login Request to the Backend Authentication Service, which validates user credentials by checking the User Database. If the credentials are valid, the system grants access and returns the Dashboard View to the Client.
2. **Product Browsing & Selection Flow:** After successful login, the Client requests the Product Catalog. The system retrieves eyewear product data from the Product Database and displays it on the UI. The user can view details, apply filters, and select products for Virtual Try-On or purchase.

3. **Cart & Order Creation Flow:** When the Client selects a product, the UI sends the selected item to the Cart Service. The system updates the cart and calculates the total amount dynamically. Upon checkout, the order details are forwarded to the Order Service, which stores the order in the database.
4. **Payment Processing Flow:** The Order Service processes the order request and stores the order details in the database. The system confirms the order using the Cash on Delivery (COD) payment method and updates the order status to Processing.
5. **System Confirmation & Completion Flow:** Once payment is confirmed, the system finalizes the order, updates inventory, and sends an Order Confirmation message to the Client. The process is then completed successfully.

The Sequence Diagram illustrates the dynamic interaction over time between the key objects and actors during the crucial task posting process.

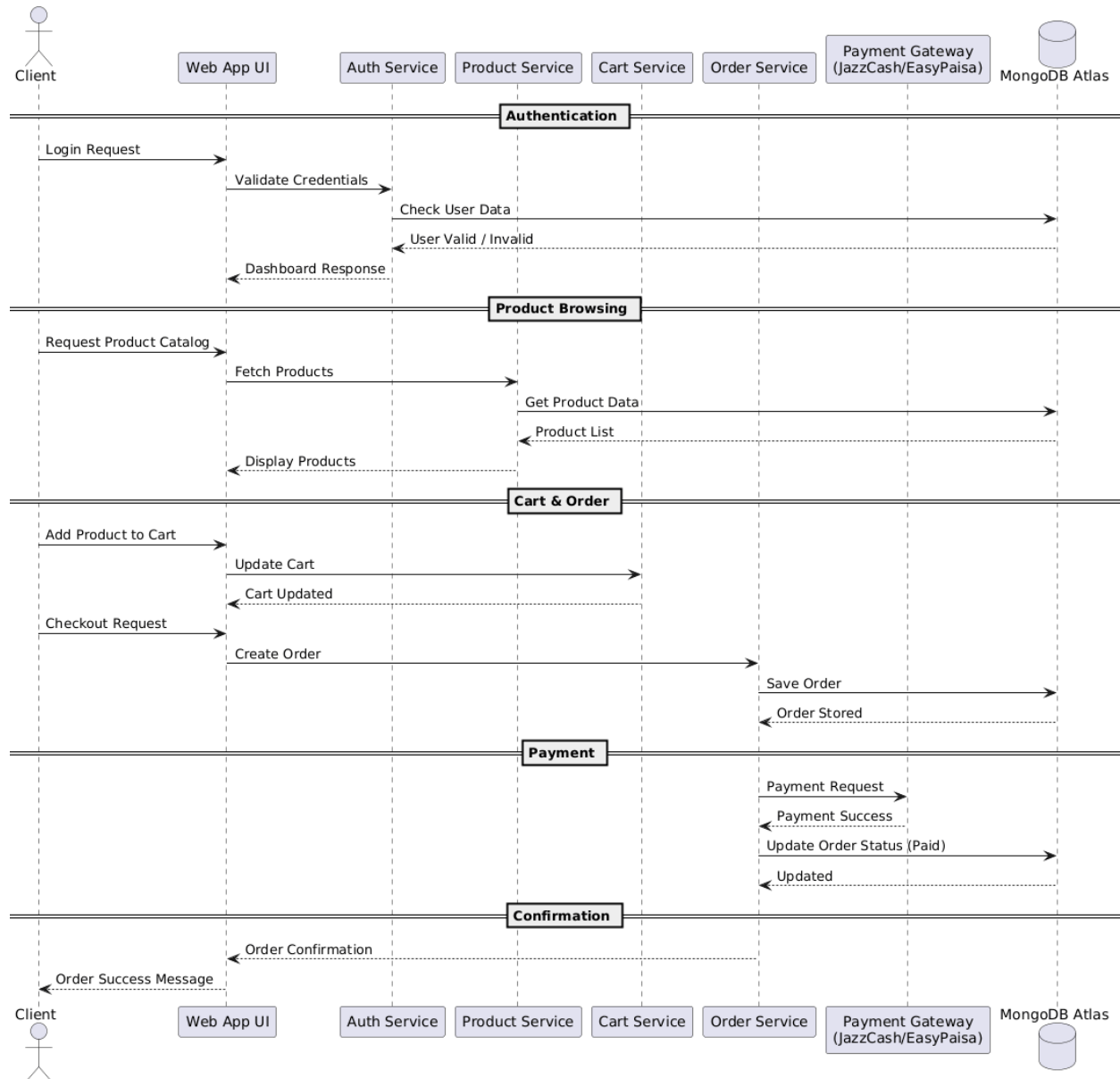


Figure 4.5 Sequence Diagram of Client / Order Processing

4.5.3 Product Posting & AI Pricing Workflow Sequence Diagram

This sequence diagram represents the complete interaction flow when a user (admin or seller-side module) adds or manages an eyewear product in the Lensify system and receives AI-based pricing suggestions. It also shows how product data is stored and how notifications are handled in the background.

1. Authentication Phase

The process begins when a user (Admin) logs into the Lensify system. The credentials are sent to the authentication service, which verifies the user from the database. After successful validation, the system generates a secure session token and grants access to the admin dashboard.

2. Product Entry & AI Pricing Phase

The admin opens the Add Product form and enters product details such as frame name, category, brand, lens type, and description.

As soon as product details are entered, the system sends relevant attributes (category, type, description) to the AI Pricing Engine.

3. Product Submission & Storage Phase

After reviewing AI suggestions, the admin finalizes the product price and submits the form. The system validates the data and stores the product in the MongoDB Atlas database under the Product collection, making it available for users in the catalog.

4. System Update & Notification Phase

- Once the product is successfully stored:
- The system updates the product catalog in real time.
- An asynchronous process is triggered.
- The Notification Service retrieves users based on preferences (face shape, category interest, browsing history).
- Notifications are sent to users informing them about new or updated eyewear products.

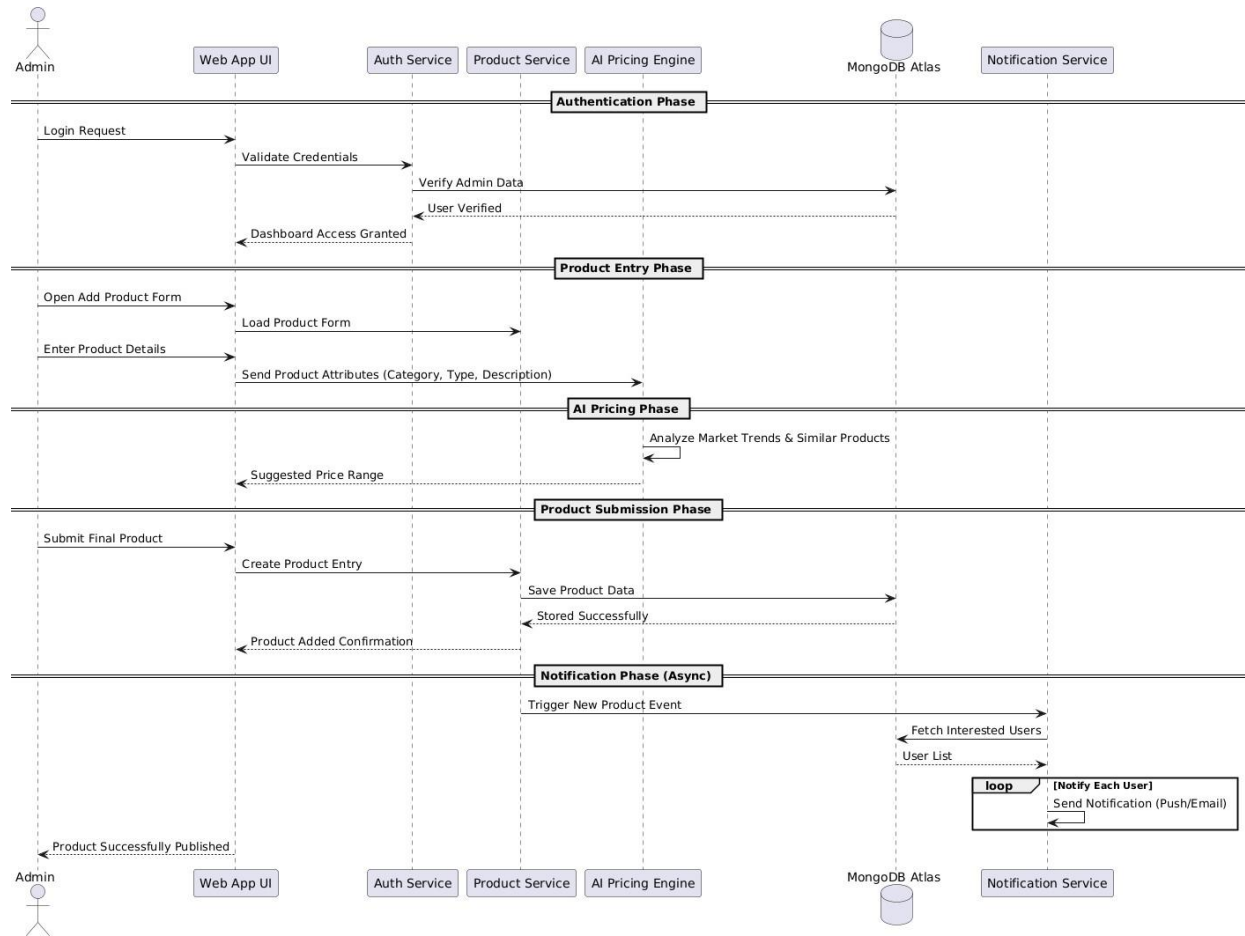


Figure 4.6 Product Posting & AI Pricing Sequence Diagram

Chapter 5

Implementation

5 IMPLEMENTATIONS

5.1 Algorithm

The implementation of intelligent features in the Lensify system is handled through dedicated AI modules integrated within the platform. These modules encapsulate all complex computational logic related to Virtual Try-On, face-shape detection, personalized recommendations, and pricing optimization. The AI components operate efficiently using client-side and backend processing (TensorFlow.js and integrated services) to ensure real-time responsiveness without degrading user experience.

The system leverages machine learning models and rule-based validation techniques to enhance decision-making, improve personalization, and ensure accuracy in critical operations such as frame selection, prescription validation, and product recommendations.

5.1.1 AI-Based Personalized Recommendation Algorithm

The AI Recommendation Algorithm in Lensify is designed to provide users with highly relevant and personalized eyewear suggestions based on their facial features, preferences, and interaction history. This algorithm plays a central role in improving user experience and increasing purchase confidence.

The process begins with face-shape detection, where the system uses facial landmark extraction (via Media Pipe Face Mesh) to classify the user's face into categories such as oval, round, square, or heart-shaped. This classification serves as the primary input for filtering suitable eyewear frames.

Next, the algorithm performs product filtering, where it selects products from the database that are tagged as compatible with the detected face shape. This ensures that only visually suitable frames are considered.

In parallel, the system analyzes user behaviour and preferences, including browsing history, selected products, and interaction patterns. This step enhances personalization by identifying user interests such as preferred styles, brands, and price ranges.

5.1.2 Dynamic Pricing Recommendation Algorithm

To enhance pricing consistency and provide better decision support, Lensify implements an AI-based Dynamic Pricing Recommendation Algorithm. This model helps in suggesting an optimal price range for eyewear products during product addition or management.

The algorithm operates using a supervised regression-based approach trained on historical product data, including attributes such as product category (glasses, sunglasses, lenses), brand value, material quality, and market demand trends.

When a new product is added, the system analyzes multiple features such as product type, frame material, brand, and comparable product pricing stored in the database. Based on this input, the AI model predicts an optimal price value.

Instead of returning a single value, the system generates a recommended price range (minimum–maximum) to provide flexibility for administrators. This range is calculated using confidence intervals around the predicted value.

5.1.3 Sentiment Analysis of Feedback

Lensify incorporates AI-based sentiment analysis to enhance product and user feedback evaluation. This feature provides deeper insights beyond simple ratings by analyzing the emotional tone of user reviews.

When a user submits a review, the system processes the textual feedback using Natural Language Processing (NLP) techniques. The algorithm classifies the sentiment into categories such as Positive, Negative, or Neutral and assigns a corresponding sentiment score.

This score is stored alongside the review data and contributes to:

- Product ranking and recommendation accuracy
- Overall system credibility
- Better understanding of customer satisfaction

The sentiment analysis module improves the reliability of feedback data and supports intelligent decision-making within the recommendation system.

5.2 External APIs

Lensify integrates multiple external APIs to enable secure transactions, AI processing, and communication features. These integrations ensure system reliability, scalability, and real-time performance.

5.2.1 Cash on Delivery (COD)

Lensify supports Cash on Delivery (COD) as its payment method. During checkout, users can place orders by providing their delivery information and confirming the purchase.

The system records order details, updates order status, and allows administrators to manage order processing and delivery. This approach provides a simple and convenient payment experience for users while ensuring accurate order tracking and management.

5.2.2 Communication and AI Services

Lensify incorporates additional APIs to support communication and intelligent system features:

- **Email Notification Service API** Used for sending automated emails such as account verification, password reset, and order confirmation.
- **AI/ML Processing Services (TensorFlow.js / Custom Models)** Used for Virtual Try-On, face detection, and recommendation engine.
- **Image Processing APIs (Face Detection / MediaPipe Integration)** Used for detecting facial landmarks and enabling real-time Virtual Try-On functionality.

Table 5.1: Details of APIs Used in Lensify

Name of API	Description	Purpose	Used In
Cash on Delivery	Built-in payment method	Order placement without online payment	Payment Module
TensorFlow.js	ML library for browser-based AI	Virtual Try-On & recommendations	AI Module
MediaPipe Face Mesh	Face detection framework	Facial landmark detection	Virtual Try-On Module
Email Service API	Email communication service	Notifications & verification	User Module

5.3 User Interface

The User Interface (UI) of Lensify serves as the presentation layer of the system, designed to provide a smooth, intuitive, and visually appealing user experience. The interface is developed using responsive web design principles to ensure compatibility across desktop and mobile browsers. It focuses on simplicity, accessibility, and seamless interaction with AI-powered features such as Virtual Try-On and personalized recommendations.

5.3.1 Authentication Views (Login, Signup)

The Login and Signup interfaces are designed with a strong emphasis on security and usability. The registration view allows users to create an account by providing essential details such as name, email, and password. Password guidelines are clearly presented to ensure strong authentication practices.

The login interface follows a clean and minimal design, enabling users to quickly access the system using their credentials. It also includes features such as password recovery and secure session handling. Upon successful authentication, users are redirected to the main dashboard, ensuring a smooth entry into the platform.

5.3.2 Product Browsing and Virtual Try-On Interface

The product browsing interface allows users to explore a wide range of eyewear products including glasses, sunglasses, and lenses. Products are displayed in an organized layout with filtering and search options based on category, price, and style.

A key feature of this interface is the integration of the Virtual Try-On system. Users can select a product and activate the try-on feature using their webcam or by uploading an image. The system overlays the selected eyewear on the user's face in real time using AI-based face detection. Additionally, AI-generated recommendations are displayed alongside products to guide users toward suitable choices based on face shape and preferences.

5.3.3 Dashboard and Order Management View

After logging in, users are directed to a personalized dashboard that acts as the central hub of the system. The dashboard provides quick access to key functionalities such as product browsing, cart management, order tracking, and recommendations.

Users can view their order history, current order status, and manage their shopping cart. The system also displays personalized product suggestions based on user activity and AI analysis, enhancing the overall shopping experience.

For administrators, the dashboard includes additional controls such as product management, inventory updates, and order monitoring, ensuring efficient system management.

5.3.4 Screenshot of Login Screen

The Lensify Login Interface follows a modern and user-friendly design. It typically features a clean layout with a visually appealing background and a centered login form for credential input. The interface includes essential elements such as email and password fields, “Forgot Password” functionality, and clear navigation options.

The design ensures ease of use while maintaining security standards. In addition, integration of support features such as AI chatbot assistance may be available to provide real-time help to users during login or navigation.

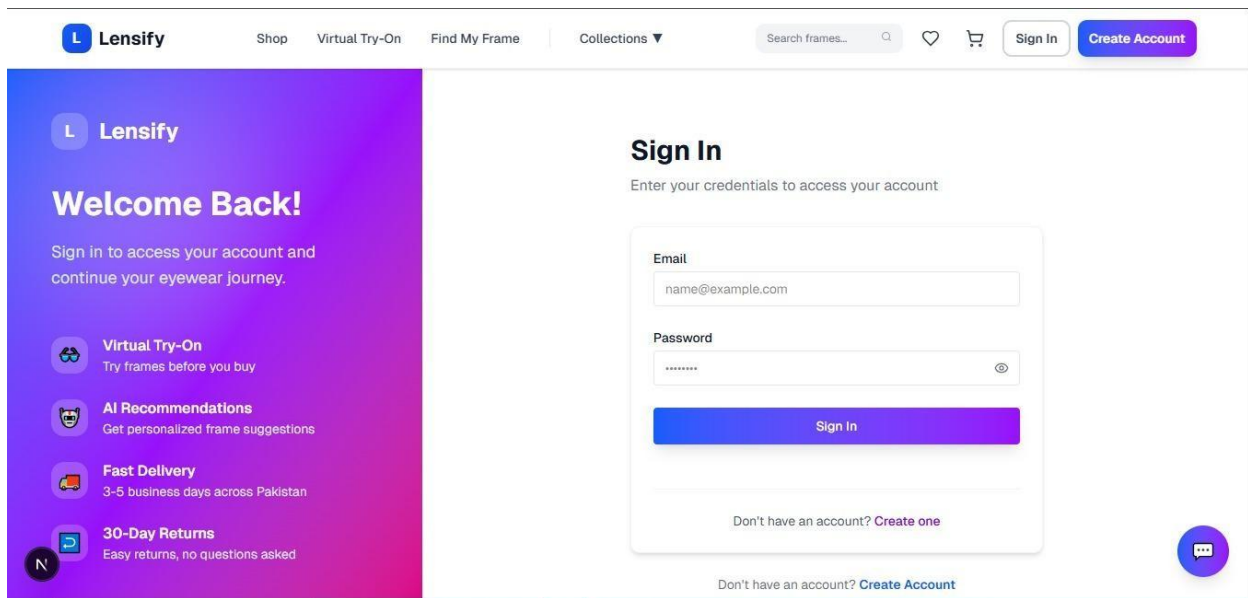


Figure 5.1 Screenshot of Login Screen

5.3.5 Screenshot of Signup Screen

The Lensify Sign Up Interface is designed to provide a smooth and secure onboarding experience for new users while maintaining a clean and modern visual layout. The interface follows a consistent design theme aligned with the platform's branding, using a minimal color scheme and a centered card-based layout to ensure clarity and focus.

The registration form captures essential user information, including Full Name, Email Address, Password, and Confirm Password. The system enforces password security requirements and provides real-time validation feedback to minimize user input errors.

To improve usability, the interface includes helper text and clear field labels guiding users on proper input formats. Navigation options are placed strategically, allowing users to easily switch between Login and Signup views.

Additionally, the interface is fully responsive and optimized for both desktop and mobile devices. Integrated support features, such as error handling messages and optional AI chatbot assistance, ensure that users can complete the registration process efficiently without confusion.

The screenshot displays the 'Create Account' page for the Lensify website. The page layout includes a top navigation bar with the Lensify logo, links to 'Shop', 'Virtual Try-On', 'Find My Frame', and 'Collections', a search bar, and 'Sign In' and 'Create Account' buttons. On the left, a purple and blue gradient sidebar promotes joining today with four key features: Personalized Experience, Premium Collection, Secure Shopping, and Order Tracking. The main content area is white and features a 'Create Account' heading with the subtext 'Sign up to start your eyewear journey'. Below this is a registration form with fields for 'First Name' (John), 'Last Name' (Doe), 'Email' (name@example.com), 'Password', and 'Confirm Password'. A 'Create Account' button is positioned at the bottom of the form. A small chatbot icon is visible in the bottom right corner.

Figure 5.2 Screenshot of Signup Screen

5.3.6 Screenshot of User Home Screen

The Lensify User Home Screen (Dashboard) serves as the central interface for user interaction, providing a personalized and intuitive overview of the platform's core features. The interface begins with a welcoming header section, offering a personalized greeting to the user, which enhances engagement and user experience. The dashboard prominently displays key sections such as featured eyewear products, AI-based recommendations, and quick access to Virtual Try-On functionality, enabling users to interact with the system efficiently. A clean, card-based layout is used to present product categories like glasses, sunglasses, and contact lenses, ensuring easy navigation and visual clarity.

The top navigation bar includes essential controls such as search functionality, cart access, order history and user profile management, allowing users to seamlessly move across different modules. The cart icon dynamically reflects added items, improving usability. Additionally, the dashboard highlights personalized suggestions generated by the AI Recommendation Engine based on the user's face shape. The Virtual Try-On feature is easily accessible from the homepage, encouraging interactive product exploration.

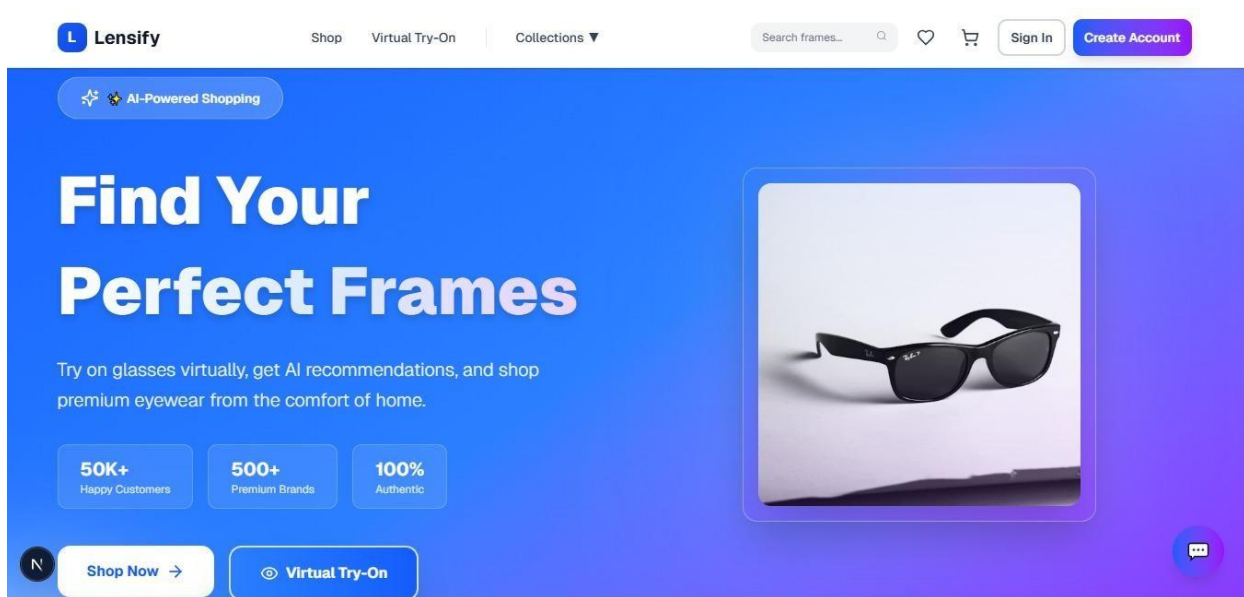


Figure 5.3 Screenshot of User Home Screen

5.3.7 Screenshot of User Analytics and Performance

This figure highlights the Analytics and Performance Visualization module within the Lensify platform, designed to provide users and administrators with meaningful insights into system usage and customer behavior. The interface is structured using a clean, card-based layout to ensure clarity and effective data interpretation.

The dashboard features two primary graphical components. The “User Engagement Trend” chart on the left utilizes a line graph with smooth transitions to visualize user activity over time, including product browsing frequency, Virtual Try-On usage, and cart interactions across a defined period. This helps in identifying peak engagement periods and overall platform usage patterns.

On the right, the “Product Preference Distribution” donut chart presents a categorized breakdown of user interests across different eyewear types such as prescription glasses, sunglasses, and contact lenses. Each category is represented using distinct color coding, allowing for quick visual comparison and easy interpretation of user preferences.

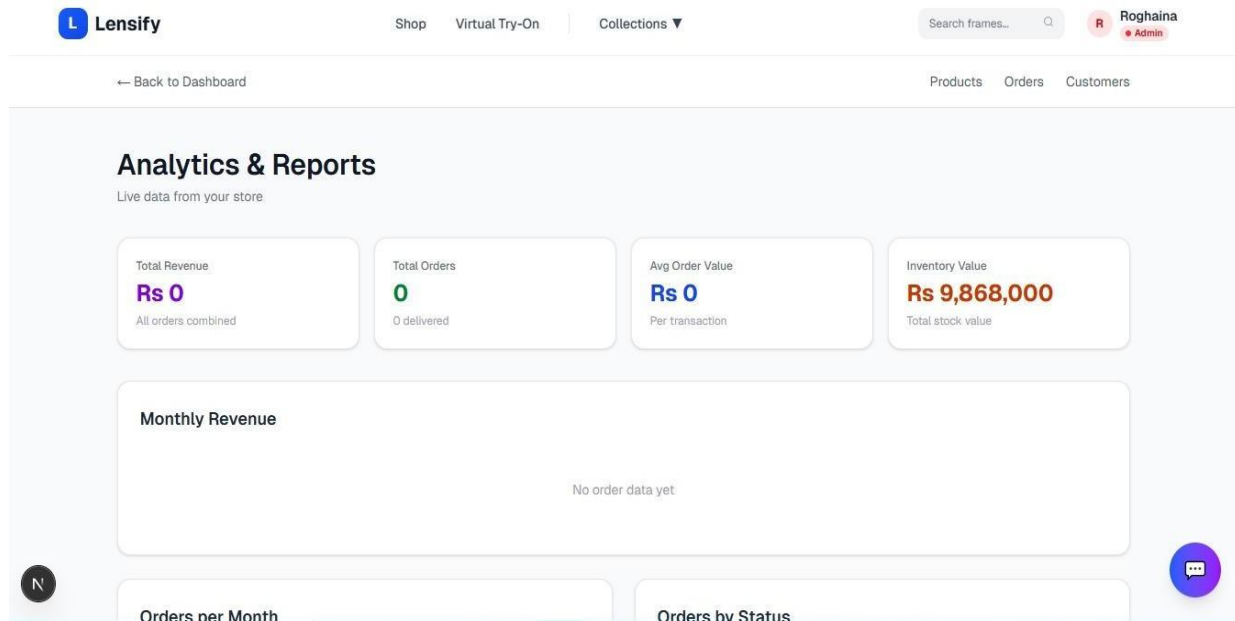


Figure 5.4 Screenshot of User Activity

5.3.8 Screenshot of User Shopping Cart

The Shopping Cart Module of Lensify is a core functional component that manages user-selected eyewear products before final purchase. It acts as an intermediary step between product selection and order checkout, ensuring users can review, modify, and finalize their choices efficiently. The module is designed to provide a smooth and intuitive shopping experience while maintaining real-time price calculation and data consistency. The interface allows users to add multiple eyewear products such as glasses, sunglasses, or contact lenses into the cart. Each cart item stores essential information including product details, quantity, selected frame type, and optional prescription data. The system dynamically calculates the total price based on selected items, applied customizations, and lens configurations. Users can update quantities, remove items, or continue shopping without losing previously selected products. The cart state is persistently managed to ensure data is retained throughout the session. Once the user proceeds to checkout, the cart data is transferred to the Order Management Module for further processing, including payment and order confirmation.

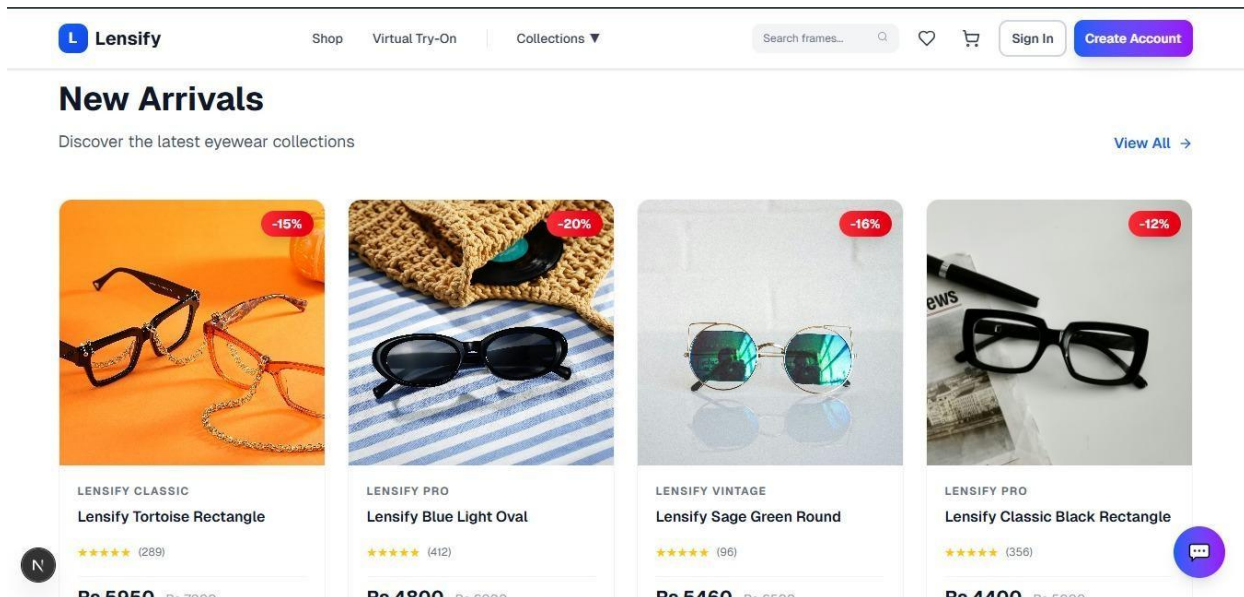


Figure 5.5 Screenshot of User Shopping Interface

5.3.9 Screenshot of Virtual Try-On (VTO)

This figure presents the Virtual Try-On (VTO) and AI Recommendation module of the Lensify platform, designed to deliver an intelligent and interactive eyewear selection experience. The interface integrates real-time face detection with personalized product recommendations to enhance user confidence during online shopping. The main section of the screen displays the Virtual Try-On window, where users can activate their webcam or upload an image to preview eyewear frames in real time. Using AI-based facial landmark detection (MediaPipe), the system accurately maps facial features and overlays selected glasses onto the user's face with proper alignment and scaling. This provides a realistic visualization of how different frames will appear before purchase. Alongside the VTO window, the AI Recommendation panel presents a curated list of suggested eyewear products based on detected face shape, previous user interactions, and style preferences. Each recommendation card includes product name, price range, and a compatibility score, indicating how suitable the frame is for the user. This helps users prioritize the most relevant options efficiently.

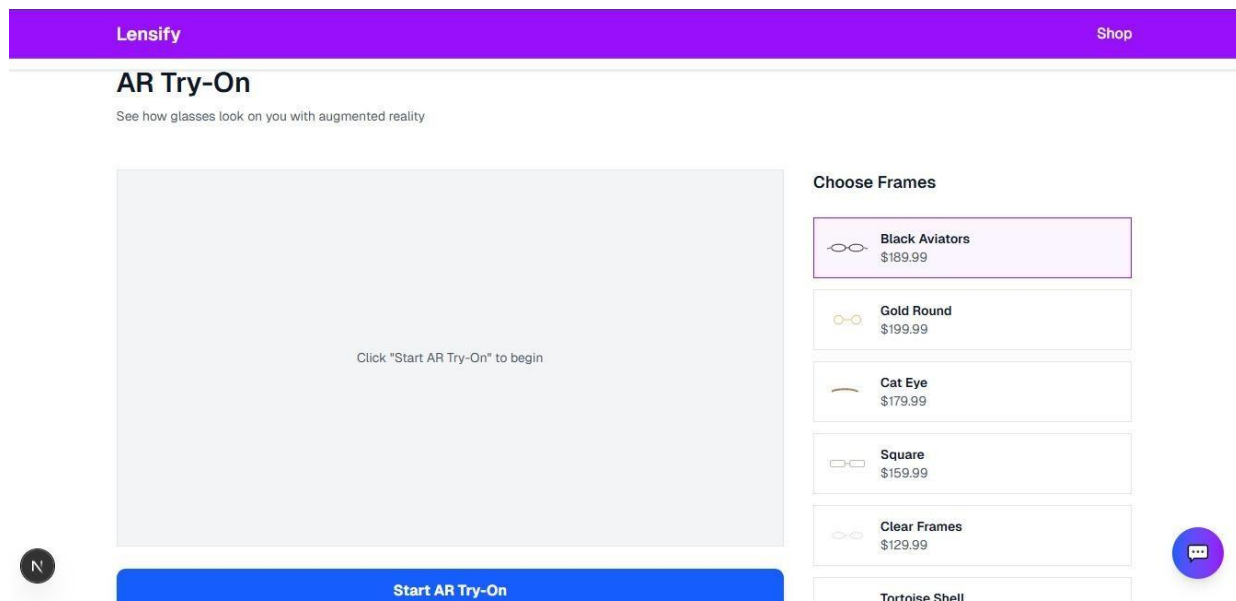


Figure 5.6 Screenshot of Virtual Try-On System interface

5.3.10 Screenshot of Admin Dashboard interface

This figure illustrates the Admin Dashboard Module of the Lensify platform, a centralized control panel designed for managing the entire e-commerce system, including users, products, orders, payments, and AI-driven services. The interface ensures efficient system monitoring, data control, and operational transparency for administrators. The main section of the dashboard provides a system overview panel, displaying key metrics such as total users, active customers, product inventory status, total orders, and completed transactions. These analytics help administrators monitor platform performance in real time. A key feature of this module is the Product & Inventory Management Panel, where admins can add, update, or remove eyewear products, manage categories (sunglasses, prescription glasses, contact lenses), and monitor stock levels to ensure availability across the platform. The interface also includes an Order Management Section, which allows administrators to track customer orders from placement to delivery. Each order record displays customer details, payment status, prescription information, and current order progress.

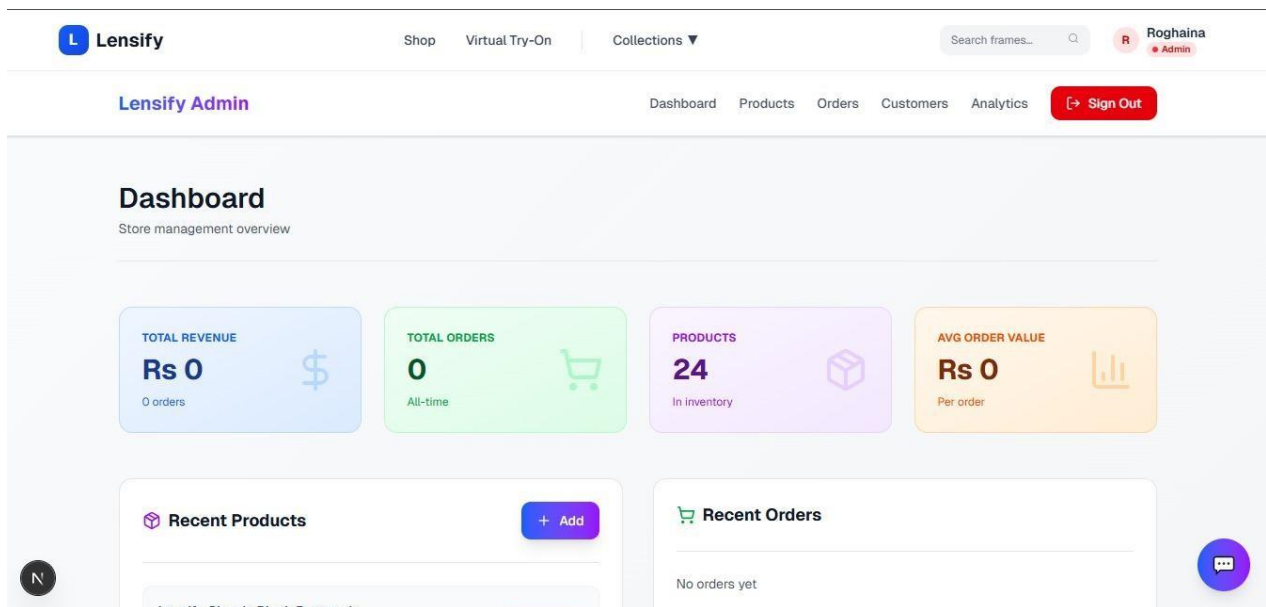


Figure 5.7 Screenshot of Admin Dashboard interface

5.3.11 Screenshot of Add to Cart System interface

This figure illustrates the Add to Cart Module of the Lensify platform, a core component of the shopping workflow that allows users to save selected eyewear products before proceeding to checkout. The interface is designed to provide a smooth, intuitive, and transparent shopping experience. The main section of the screen displays the Cart Interface, where users can view all selected eyewear products in a structured list format. Each cart item includes product image, frame name, lens type (if applicable), quantity selector, and individual price details, allowing users to manage their selections easily. A key feature of this module is the Dynamic Price Calculation Panel, which automatically updates the total cost in real time based on product selection, quantity changes, and additional customization such as prescription lenses or frame upgrades. This ensures accurate billing before payment.

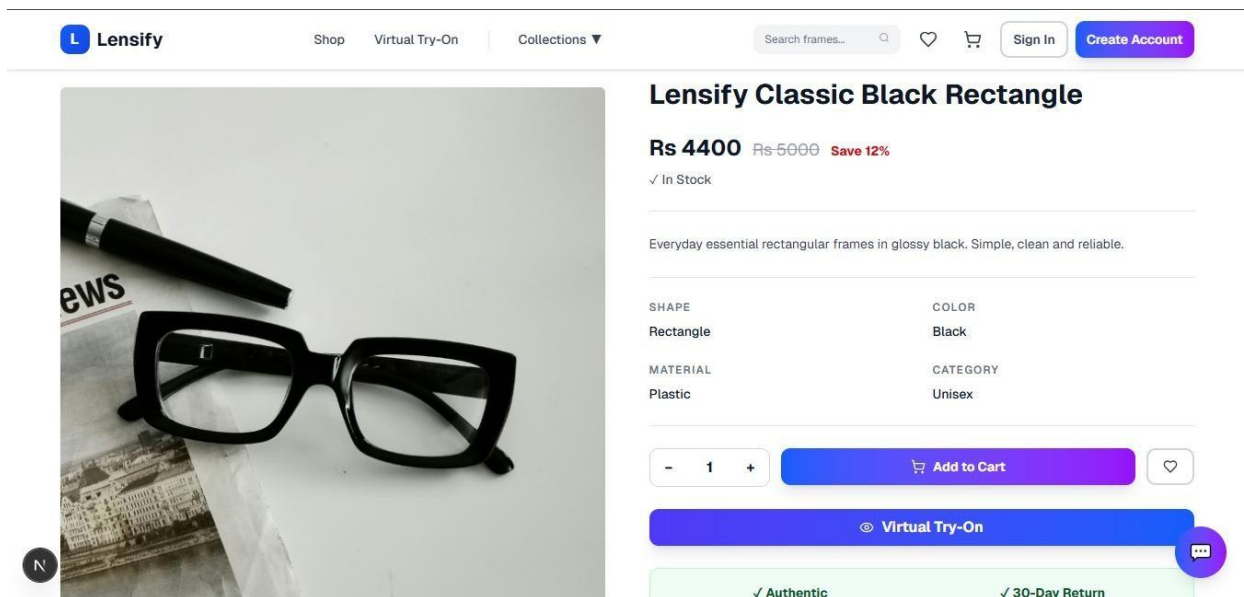


Figure 5.8 Screenshot of Add to Cart System interface

5.3.12 Screenshot of AI Face Recommendation Interface

This figure illustrates the AI Face Recommendation Module of the Lensify platform, which plays a central role in delivering a personalized eyewear shopping experience. The interface is designed to intelligently analyze the user's facial features and recommend the most suitable eyewear based on face shape and visual compatibility. The main section of the screen displays the Live Face Detection Interface, where the user activates the camera or uploads an image. The system utilizes advanced AI models (Media Pipe Face Mesh and TensorFlow.js) to detect facial landmarks and accurately identify the user's face shape, such as round, oval, square, or heart-shaped.

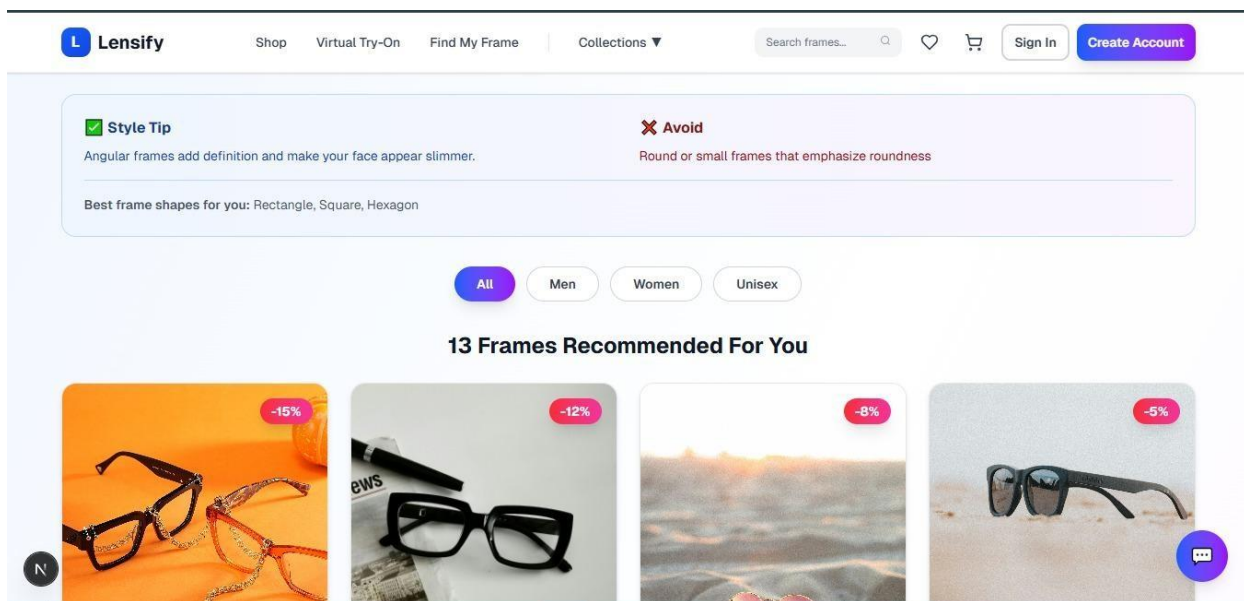


Figure 5.9 AI Face Recommendation Interface

Chapter 6

Testing and evaluation

6 TESTING AND EVALUATION

6.1 Manual Testing

Manual testing was performed across the complete Lensify platform to verify system functionality, AI feature accuracy, UI responsiveness, and integration consistency. The testing ensured that all modules behave according to the defined system requirements and design specifications.

6.1.1 System Testing

System testing was conducted on the fully integrated Lensify application to validate that all modules (Frontend, Backend, AI Services, and Database) operate correctly in a unified environment. The objective was to ensure proper communication between components such as product management, Virtual Try-On, AI recommendations, cart operations, and payment processing.

Table 6.1: System Testing

No.	Test Objective	Test Script	Expected Result	Result
1	Verify complete user workflow (Browse → Try-On → Cart → Payment)	User selects product → uses VTO → adds to cart → completes payment	All modules execute sequentially without error and order is successfully placed	Pass
2	Verify cross-device responsiveness	Access Lensify on mobile, tablet, and desktop browsers	UI remains responsive, consistent, and functional	Pass
3	Validate AI module independence	Run Virtual Try-On while browsing products	AI processing runs smoothly without affecting UI performance	Pass

6.1.2 Unit Testing

Unit testing focused on isolating and validating the smallest testable components, such as individual functions, data model methods, and security logic.

Unit Testing 1: Authentication Unit Testing

Unit testing focused on validating individual components such as authentication, cart logic, AI recommendation functions, and payment validation.

Table 6. 2: User Authentication

No.	Test Case	Input	Expected Result	Result
1	Valid customer login	Correct email & password	Redirect to user dashboard	Pass
2	Invalid password login	Correct email, wrong password	Display error message	Pass
3	Role-based access validation	Customer login	Correct dashboard loaded based on role	Pass

Unit Testing 2: AI Dynamic Pricing Model Integrity

To ensure that the AI-based pricing recommendation system in Lensify generates accurate, realistic, and market-aligned price suggestions for eyewear products based on product type, lens customization, and user requirements.

Table 6. 3: AI Rrecommendation &VTO Testing

No.	Test Case	Input	Expected Result	Result
1	Virtual Try-On activation	Face image / webcam input	Glasses correctly overlay on face	Pass
2	Recommendation accuracy	User preference + face shape data	Relevant eyewear suggestions generated	Pass
3	AI performance validation	Multiple product requests	Fast response without delay	Pass

6.1.3 Functional Testing

Functional testing was performed to verify that all core features of the Lensify platform operate correctly according to the defined functional requirements. The testing mainly focused on AI-driven modules (Virtual Try-On, Recommendation System), communication system, and transactional workflows to ensure system reliability and correctness.

Table 6.4: Functional Testing

No.	Test Case	Attribute and Value	Expected Result	Result
1	Verify AI Recommendation System for eyewear products	User searches: “glasses for round face with blue light protection”	System analyzes face shape + preferences and returns top relevant eyewear products ranked by AI recommendation score	Pass
2	Test Virtual Try-On (VTO) integration with product selection	User activates camera and selects any eyewear frame	System detects facial landmarks using AI (MediaPipe/TensorFlow.js) and overlays selected eyewear accurately in real time	Pass
3	Validate AI Chatbot Response System	User types question in chat widget and submits message	User message is sent to AI chatbot (Llama 3.2 via Ollama API) and response is returned instantly through REST API call	Pass

6.1.4 Integration Testing

Integration testing was conducted to ensure smooth interaction between frontend, backend services, AI modules, and external APIs. It verifies that all system components work together as a unified e-commerce platform.

Table 6.5: Integration Testing

No.	Test Case	Attribute and Value	Expected Result	Result
1	End-to-End Product Flow with AI Recommendation	User selects eyewear → AI suggests similar products → item added to cart	AI Recommendation Engine successfully interacts with Product Service and updates suggested products in real time	Pass
2	Virtual Try-On + Product Selection Integration	User applies VTO while browsing products	Selected product is visually rendered on user face and linked correctly with product ID in backend cart system	Pass
3	Cart to Payment Integration	User adds items to cart and proceeds to checkout	Cart data is correctly transferred to Payment Module and total amount is calculated with prescription/customization cost	Pass
4	Admin Product Management Integration	Admin updates product price or stock	Changes are instantly reflected in frontend product listing and database consistency is maintained	Pass

6.2 Automated Testing

Automated testing ensures the long-term stability, scalability, and reliability of the Lensify platform. It is integrated into the CI/CD pipeline so that every code update is validated before deployment, preventing system regression and errors in AI modules or payment systems.

Table 6.6: Automated Testing

Tool Name	Tool Description	Applied On	Results
Manual API Testing (Postman)	REST API testing tool used to validate all backend endpoints	Authentication, Product APIs, Cart, Orders, AI Chatbot endpoints	All API endpoints return correct responses with proper status codes
TensorFlow.js Testing Mode	Used for validating AI model behavior in Virtual Try-On and recommendation engine	Face detection, eyewear overlay, recommendation accuracy	Confirms AI model accuracy and real-time performance
Manual Order Testing	Validates Cash on Delivery workflow from cart to order confirmation	Payment & Checkout Module	Order placed and confirmed successfully
GitHub Actions CI/CD Pipeline	Automated build, test, and deployment workflow	Entire system (Frontend + Backend + AI modules)	Ensures continuous integration and prevents deployment of faulty code

Chapter 7

Conclusion

7 CONCLUSION AND FUTURE WORK

7.1 Conclusion

The development of Lensify successfully achieved its primary objective of designing and implementing an intelligent, user-friendly, and AI-powered eyewear e-commerce platform that enhances the online shopping experience through Virtual Try-On and personalized recommendations. The system was developed using a modular, layered architecture, ensuring scalability, maintainability, and efficient integration of AI services. By applying an iterative development approach, all core modules were successfully implemented and tested.

The major achievements of the Lensify system include:

- **AI-Powered Virtual Try-On (VTO):** The system successfully integrates MediaPipe Face Mesh and TensorFlow.js to detect facial landmarks and overlay eyewear in real time, allowing users to virtually try glasses before purchase.
- **AI Recommendation System:** A smart recommendation engine analyzes face shape, user preferences, and product metadata to suggest the most suitable eyewear products, improving personalization and decision-making.
- **Dynamic Product and Cart Management:** The platform provides a smooth shopping experience through real-time product browsing, cart management, and checkout functionality with accurate price calculations.
- **Secure Payment Integration:** Cash on Delivery payment method is implemented, which is the most widely used payment method in Pakistan, ensuring safe and reliable order processing.
- **Admin Control and System Management:** The admin module ensures effective management of products, users, and orders, maintaining system integrity and operational control.

7.2 Future Work

Although Lensify fulfills all current functional requirements, further enhancements can significantly improve its intelligence, scalability, and user experience.

1. **Advanced AI Recommendation Enhancement:** Future versions can incorporate deep learning models to improve recommendation accuracy by analysing user behaviour patterns, purchase history, and seasonal trends.
2. **3D Virtual Try-On Enhancement:** The current 2D overlay system can be upgraded to a full 3D rendering engine, allowing more realistic eyewear visualization from multiple angles and improved depth accuracy.
3. **Mobile Application Development:** A dedicated Android and iOS application can be developed to increase accessibility and improve performance for mobile users, especially in low-bandwidth regions.
4. **AI-Based Face Shape Classification Improvement:** The system can be enhanced with a more advanced facial classification model to improve precision in detecting face types and recommending better-suited eyewear.
5. **Augmented Reality (AR) Integration:** Future implementation of AR-based try-on using smartphone cameras can provide a more immersive shopping experience.
6. **Multi-Vendor Marketplace Expansion:** Lensify can evolve into a multi-seller platform where different optical stores can register and sell products, expanding inventory and market reach.
7. **Enhanced Customer Support System:** Integration of an AI chatbot with natural language understanding can provide 24/7 assistance for product selection, order tracking, and support queries.

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